COSC-3370 Software Engineering

SMRT Vocab

November 19, 2024

Lang Gang

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Configuration Manager – Jacob Poland

Test Manager – Zachary Kao

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Design Manager - Garrett Strader

Presentation Outline

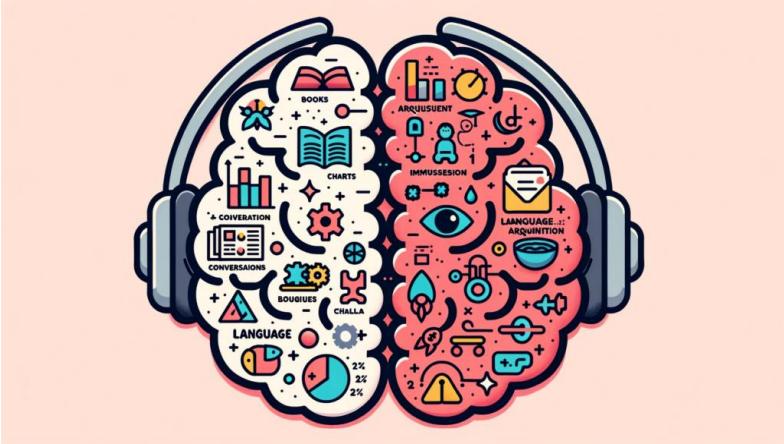
- Project Vision
- Project Planning
- Project Requirements
- Product Design
- Project Implementation
- Product Testing
- Lessons Learned
- Product Launch / Demonstration
- Questions

Project Vision

- Vocabulary flashcard application
- Efficiently progress users through vocabulary
- Language-learners, educators, hobbyists, and researchers
- No gamification

Project Vision

- **Goal**: Sell application, provide API rights, linguistics
- SMRT Vocab supplements and minimizes vocabulary efforts
- Break down the "vocab problem" - solve it.
- Users can spend less time not knowing
- More time enjoying their language of choice

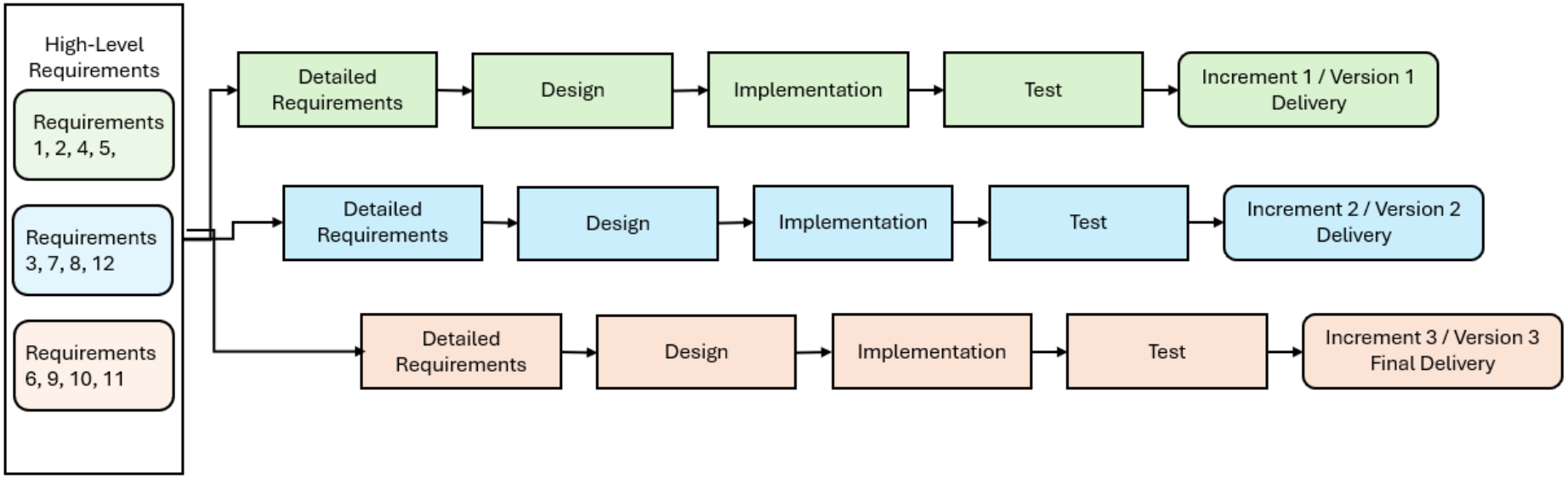


Project Planning

- Communication and Meetings
- Development Tools
- Risk

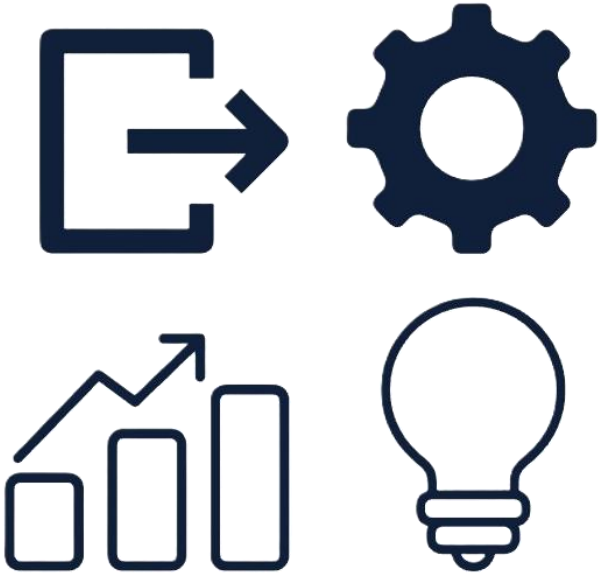
Project Planning

- Incremental Software Development Life Cycle Model (SDLCM)



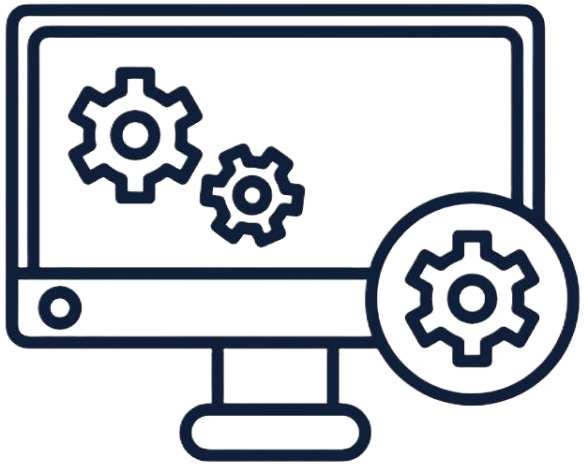
Functional Requirements

- User Management and Progress Tracking
- Flashcard Interaction and Study Modes
- Feedback and Learning Support
- Navigation and Settings



Hardware Requirements

- Minimum 8 GB RAM
- Minimum 1 GB of available storage
- Minimum quad core processing support
- Monitor resolution of 1280 x 1024 px

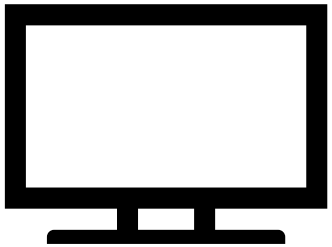


Quality Requirements

- Performance and Responsiveness
- Accessibility and Inclusivity

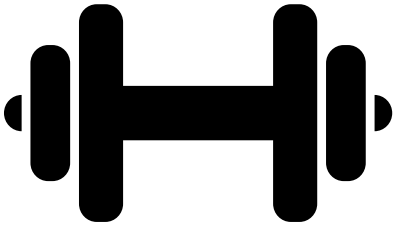
Product Design

- Teaches vocabulary, not grammar
- Facilitates an intuitive learning environment
- Intended to be paired with media of foreign language



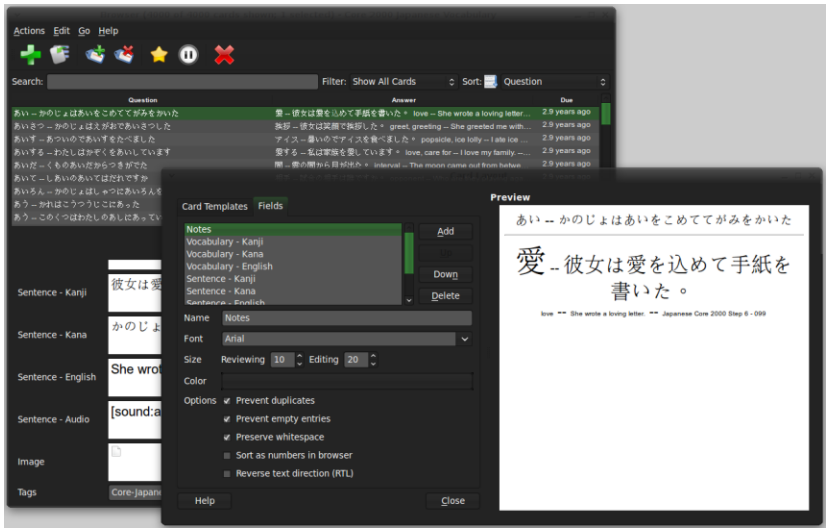
Product Design

- Algorithm does the heavy lifting
  - Tracks progress and accuracy
  - Maintains window of pertinent words per user

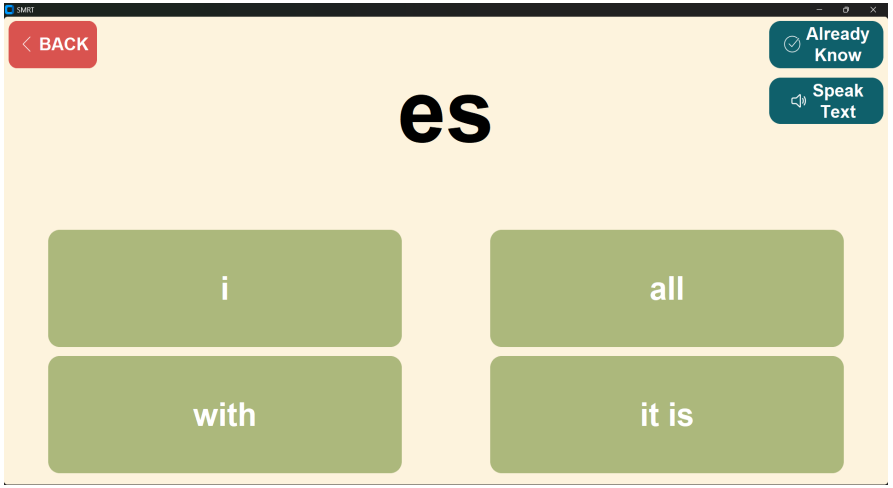


Simple is Beautiful

- Unlike our competitors, we strive to make the UI easy to understand



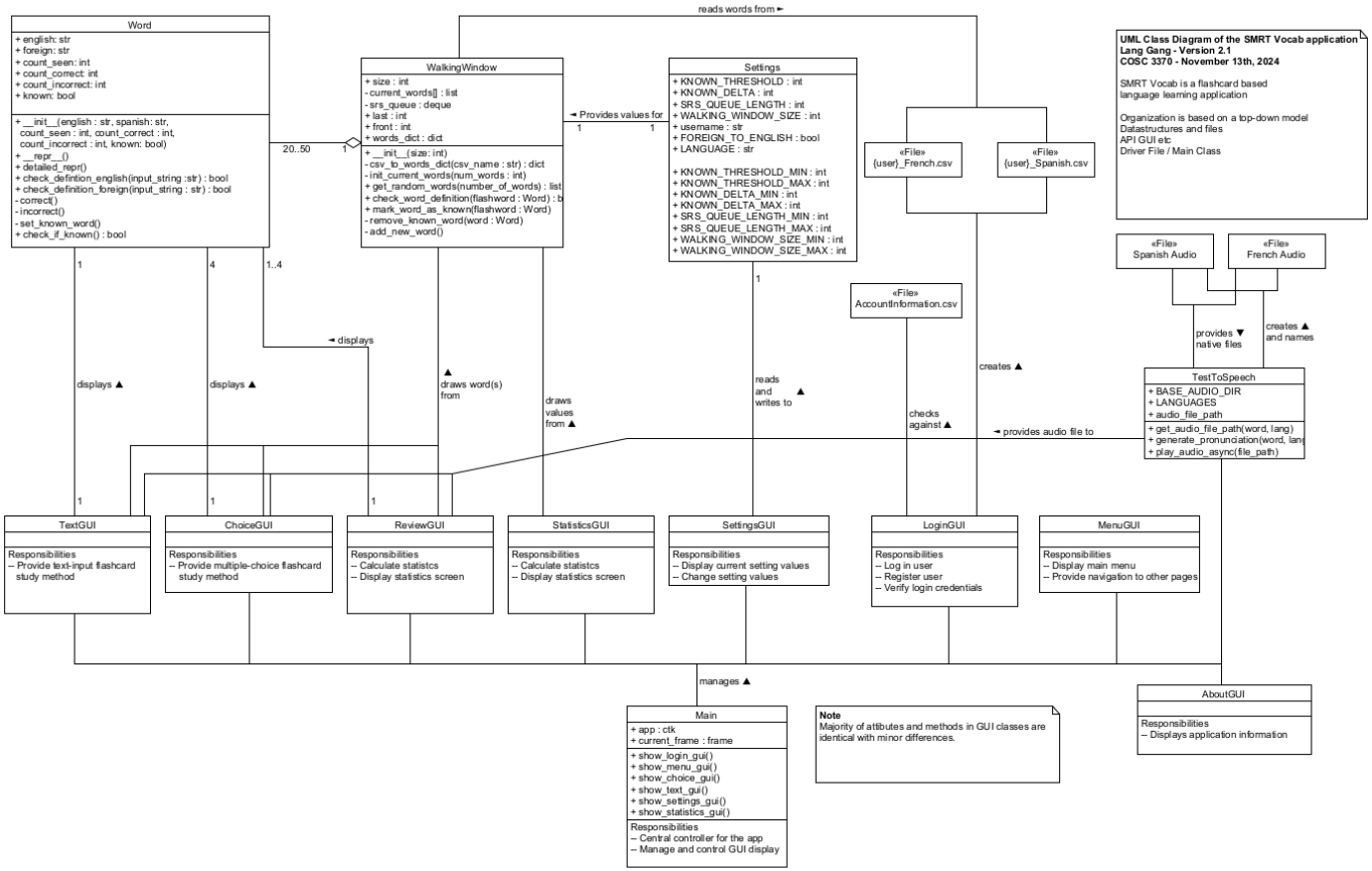
Anki



SMRT

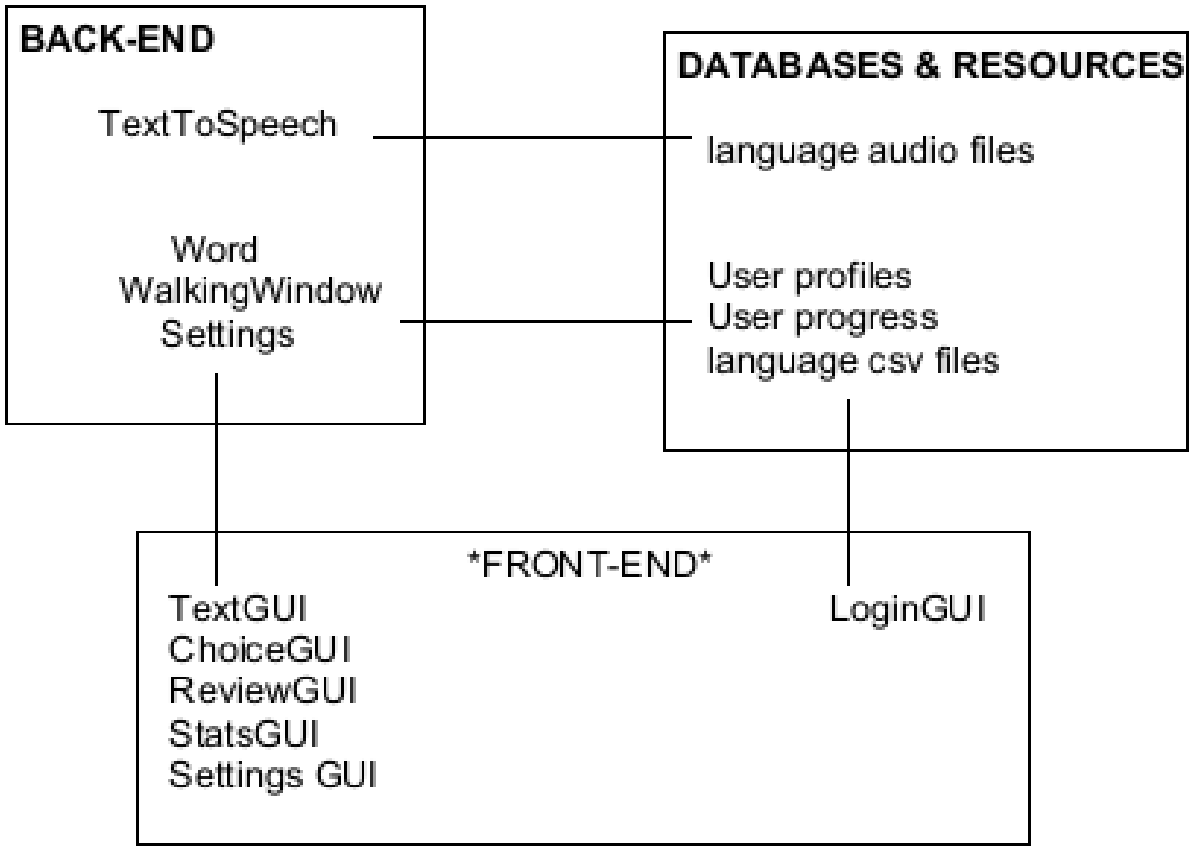
Product Design

➢ Architecture



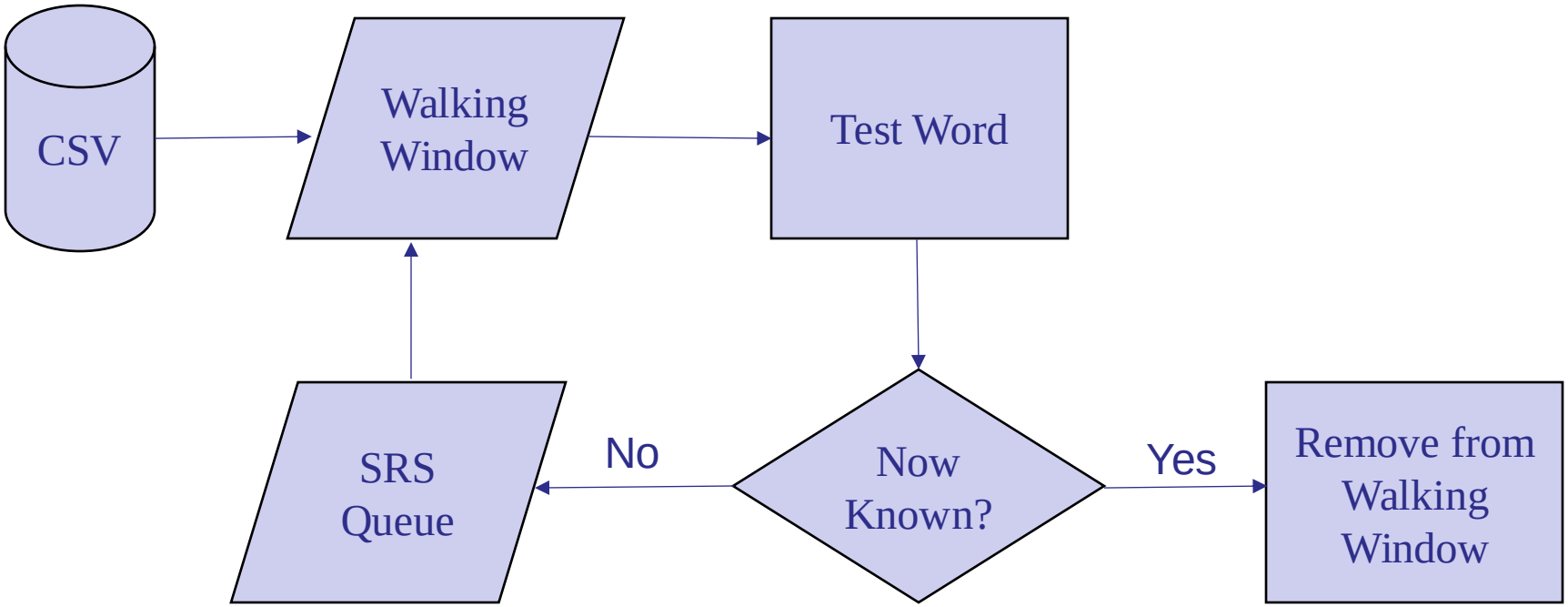
Product Design

- Architecture - Simplified



Product Design

- Lifecycle of a Word



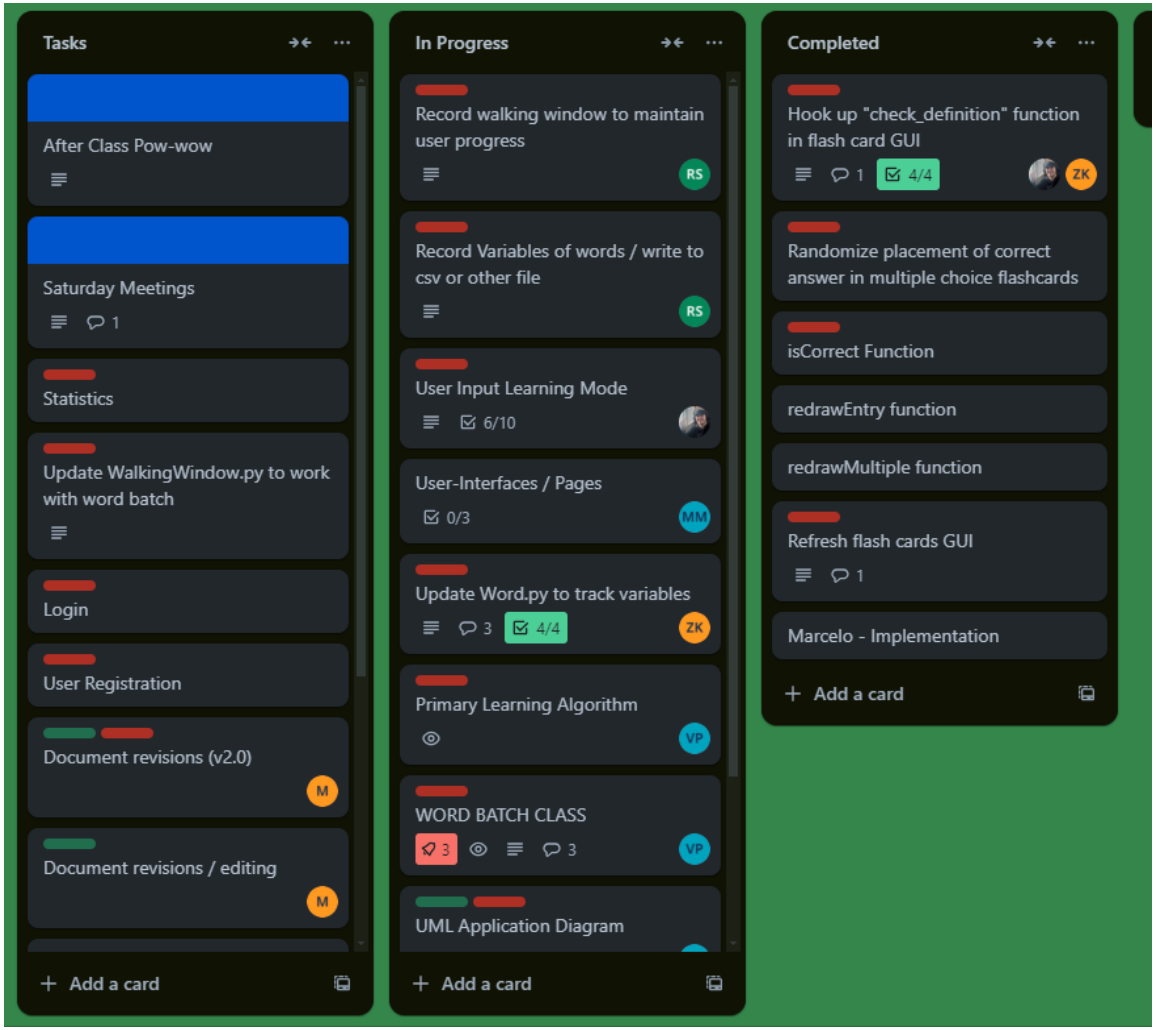
Project Implementation

- **Technology Stack:**
  - Python
  - Tkinter, CustomTkinter
  - Pygame, Pillow, GoogleTTS

Project Implementation

- **Teamwork**
  - **Planning and Development:** Weekly meetings
  - **Collaboration Tools:** Trello, Discord, GitHub

Project Implementation - Trello



Product Testing

- TC-3 and TC-4 : Flashcard Testing
  - Flashcard studying feature
  - Text-to-Speech feature
  - Already Known feature
  - Data Persistence
- ~12 steps, at most 30-minute tests

Product Testing

- **Successful Testing**
  - Logging for easy debugging
  - Rigorous testing
  - Bugs are reported, reproduced, and fixed
- **Challenges**
  - Issues adding Arabic language
  - CSV encoding caused multiple issues

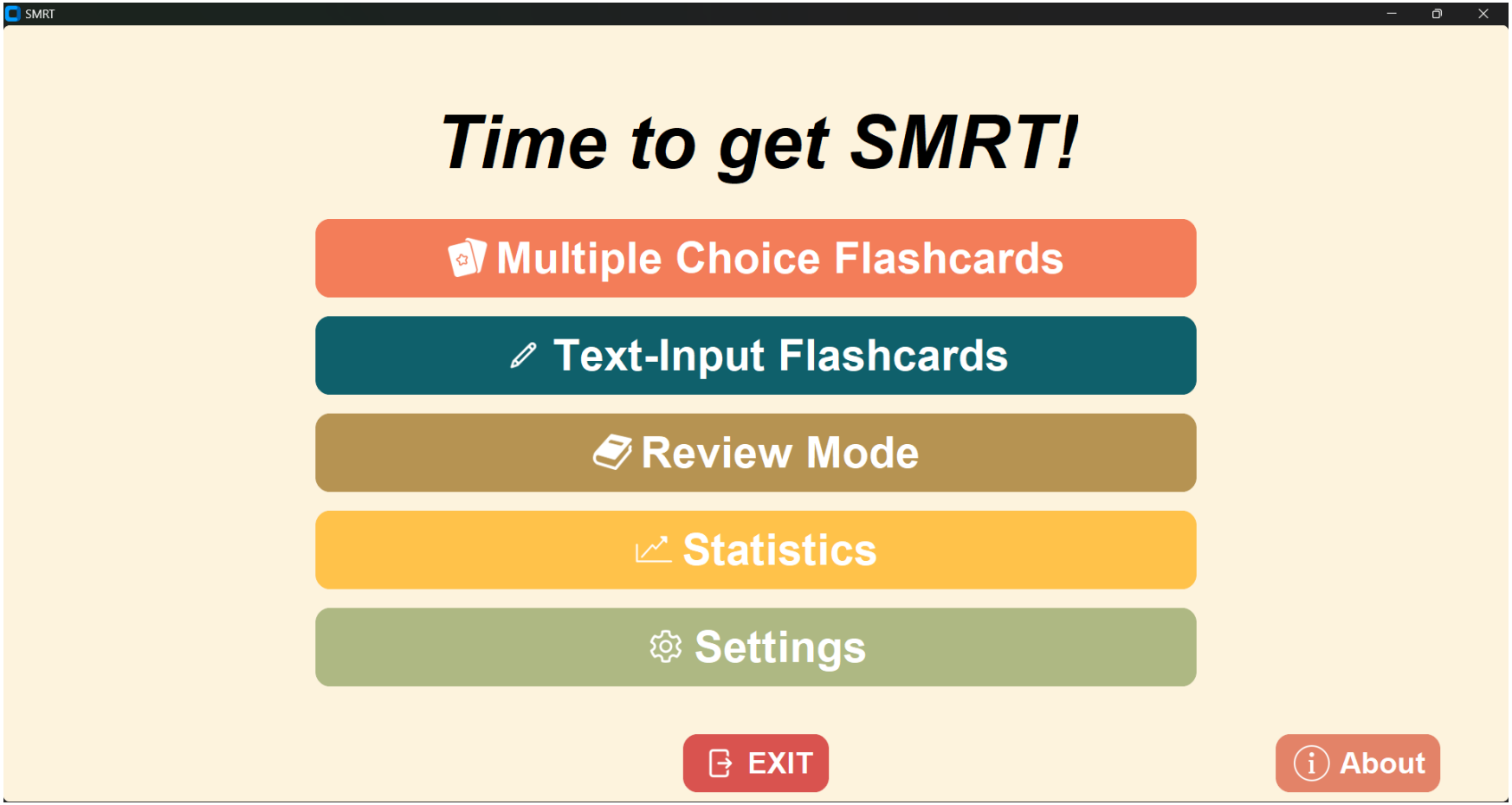
Lessons Learned

- **Challenges**
  - Onboarding a new member mid-project
  - Scheduling for a large team
  - First time coding in Python for a large project
- **Motivating Factors**
  - Eager and communicative members
  - Proactive members developed prototype early on

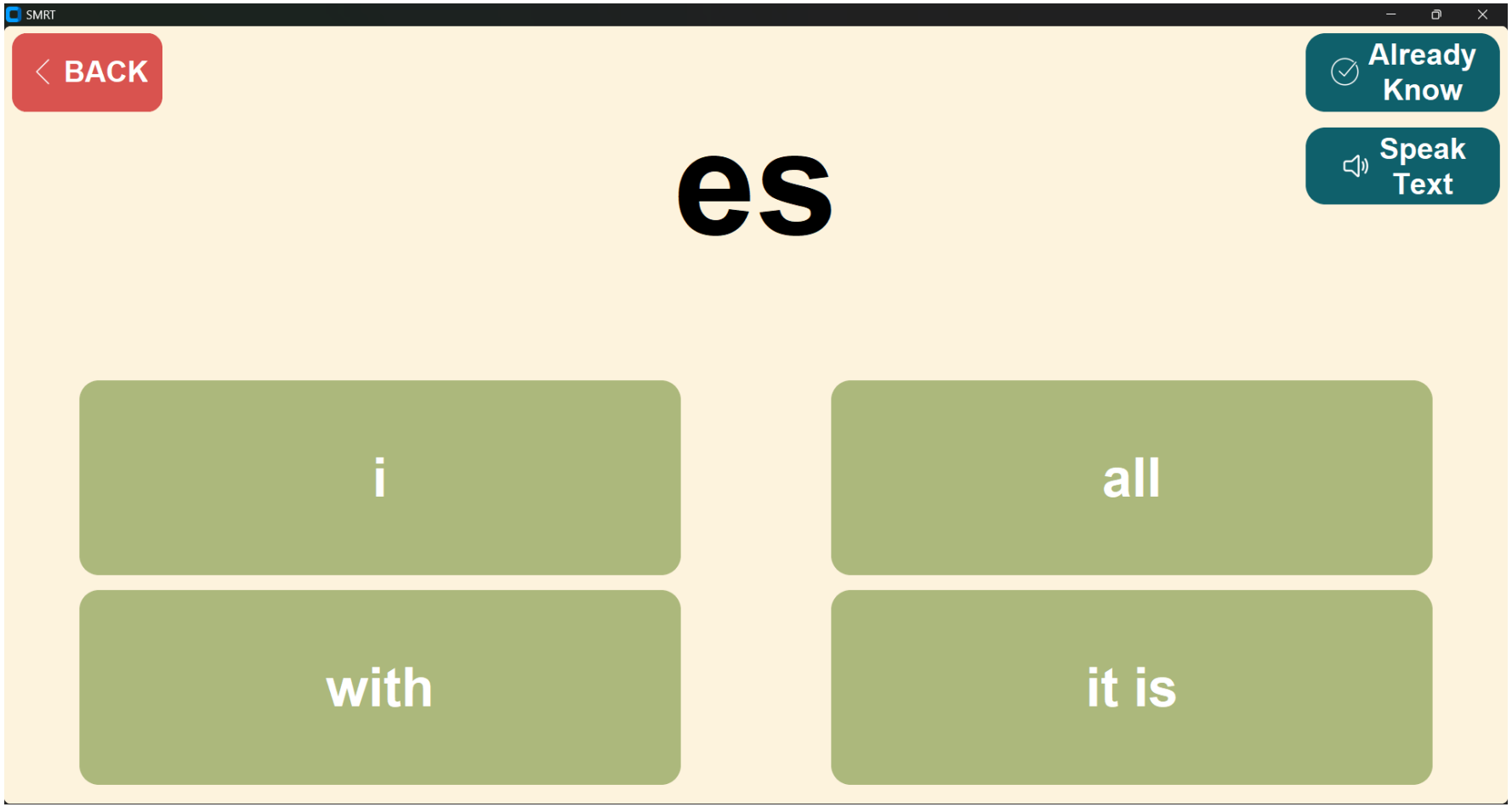
Lessons Learned

- **Successes**
  - Communication is frequent and regularly occurring
  - Shared vision across the team
  - Team of motivated individuals
- **Improvements**
  - Earlier adoption of project management tools
  - Improved scheduling for meetings

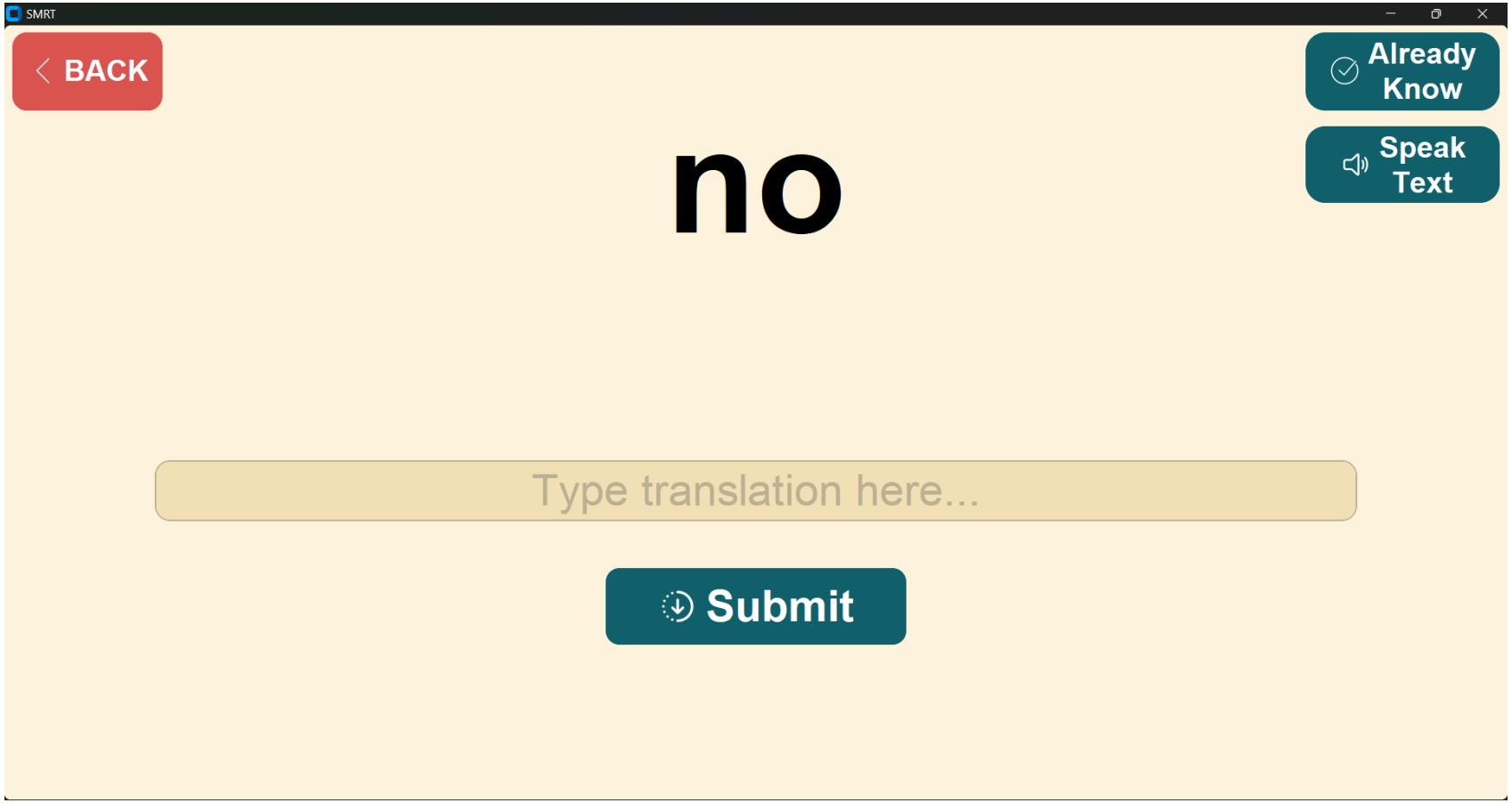
Product Feature – Main Menu



Product Feature – Multiple Choice Flashcards



Product Feature – Text-Input Flashcards



Product Demonstration

- Login
- Register
- Flashcards
- Settings

Questions

Check us out on GitHub!

https://github.com/Inhuman-creation/SMRT-PROJECT

Lang Gang

SMRT Vocab
Project Vision Document

Version 3.0
11/21/24

Release Checklist

Checklist Item	**Control (Yes/No)**
The document has passed the quality assurance process.	**Yes**
The document is visually appealing and has a professional look and structure.	Yes
All placeholders such as text within <> are replaced with appropriate text.	Yes
The document is prepared in the "Justify Text" format.	Yes
The document does not include typos, different fonts, and misalignments.	Yes
In the revision history, the first version states "The first version". Additional versions include an entry highlighting all the changes from the previous version.	Yes
The table of contents (TOC) is updated as necessary.	Yes
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	Yes
The language grammar is used appropriately.	Yes
Sentences are written as full sentences.	Yes
The wording in the document is formal.	Yes
The wording in the document is clear.	Yes
The wording in the document is professional.	Yes
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	Yes
The team member names are specifically spelled out in the document preparation and approvals sections.	Yes
The purpose and scope of the document are adequately described.	Yes
All sections include the necessary and adequate information.	Yes
All stakeholders and users are identified as related to the product.	Yes
The user environment is adequately analyzed and described.	Yes
The product alternatives and competitive products are adequately analyzed.	Yes
The "Precedence and Priority" section includes the list of features in a priority order.	Yes

Final Control	Yes/No
The document is checked based on the "Team Project Artifacts Common Errors" Guideline. The document is ready for release.	**Yes**
The document is not ready for release.	No

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/2024
Zachary Kao	Project Test Manager	Zachary S Kao	11/21/2024
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/2024
Voss Purkey	Project Manager	Voss Purkey	11/21/2024

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Garrett Strader	Project Design Manager	Garrett Strader	11/21/2024
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/2024
Zachary Kao	Project Test Manager	Zachary S Kao	11/21/2024
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/2024
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/2024
Marcelo Barahona	Software Engineer	Marcelo Barahona	11/21/2024
Voss Purkey	Project Manager	Voss Purkey	11/21/2024

Revision History

Date	Version	Description	Author
9/10/24	1.0	The first version.	Voss Purkey
9/20/24	2.0	Added new team member.	Voss Purkey
9/25/24	2.1	Added user registration and login to sections 5 and 8.	Jacob Poland
10/21/24	2.2	Corrected on issues C13 and C6 for QA. Corrections made to document introduction. Deleted developers from Stakeholder table. Organized and expanded on 4.2. Re-organized section 8 to provide a more logical priority of features. Wording.	Voss Purkey
11/10/24	2.3	Revisions for the updated project as it stands to date.	Ryleigh Sibley
11/11/24	2.4	Modified the table of contents	Voss Purkey
11/14/24	2.5	Updated section 3.4 Apply justified formatting.	Zachary Kao
11/21/24	3.0	Final check for submission.	Malak Mahdy Marcelo Barahona

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Project Vision Document

1. Introduction

1.1 Purpose of the Document

The purpose of this document is to define and describe the SMRT Vocab application. It provides a description of the product features, the business environment and positioning of the product as well as serving as a source document for future documents within the product's development lifecycle. This document discusses the scope, goals, stakeholders, and user analyses for SMRT Vocab. It also provides a brief description of product requirements, and priorities of quality and features. The inspiration and basis for the application and how it fits into language learning pedagogy are provided as references and product descriptions.

1.2 Scope of the Document

This document outlines the strategic framework and overarching goals for the development and deployment of the software. It defines the project's objectives, target audience, and key functionalities, detailing how the software will address specific needs or problems within its intended context. This document serves as a guiding reference, specifying the project's deliverables, timelines, and constraints while providing a shared understanding of the project's purpose and expected outcomes. It also includes an overview of the project's key features, technological requirements, and success criteria, ensuring all team members are on the same page regarding the vision, scope, and ultimate impact of the software program.

1.3 References

1. Lang Gang (2024). Internal Document: Project Proposal PowerPoint. Point of Contact for this document: Voss Purkey – Project Manager.
2. Zipf, G. K. (2013). The psycho-biology of language: An introduction to dynamic philology. Routledge
3. SUZANNE HAZENBERG, JAN H. HULSTUN, Defining a Minimal Receptive Second-Language Vocabulary for Non-native University Students: An Empirical Investigation, Applied Linguistics, Volume 17, Issue 2, June 1996, Pages 145–163, https://doi.org/10.1093/applin/17.2.145
4. Hu, Marcella & Nation, Paul. (2000). Unknown Vocabulary Density and Reading Comprehension. Reading in a Foreign Language. 13. Accessed 11092024: https://www.researchgate.net/publication/234651421_Unknown_Vocabulary_Density_and_Reading_Comprehension
5. Multiple Authors (1997). Vocabulary: Description, Acquisition and Pedagogy. Cambridge University Press. Accessed 11092024: https://catdir.loc.gov/catdir/samples/cam031/97042211.pdf
6. Shabani, Karim & Mohammad, Khatib & Ebadi, Saman. (2010). Vygotsky's Zone of Proximal Development: Instructional Implications and Teachers' Professional Development. English Language Teaching. 3. DOI: 10.5539/elt.v3n4p237.

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Statistically Motivated Repetition Trainer.	SMRT	Name of project/program.
Comprehensible Input	CI	Language learning technique that uses a base knowledge and includes unknown words amended by context via images or other known words. This is close to methods such as the direct method or the natural method.
Spaced Repetition System	SRS	The name given to the system of smarter flashcard inspired learning which considers the frequency of views of a point of knowledge, the level of ease it is learned, and the number of times needed to prove a user knows it.
Personally Identifiable Information	PII	Refers to any data that can be used to identify an individual, such as names, addresses, Social Security numbers, or biometric records.
Zipf's Law	-	Statistical law that reappears in various places. In linguistics this law describes the positive relation between a word's frequency and the sequential order of the following words.
Pareto Principle	-	Business principle described by Pareto as the 80/20 rule. It is associated with Ziph's law. An example would be that 80% of the effort in language learning accounts for 20% of the success or 20% of employees do 80% of the work.
Zone of Proximal Development	-	Pedagogical principles are referenced in various language learning methods. The principle states that people learn automatically via context and understanding when they understand around 98% of the content. This is associated with CI.

2. Positioning

2.1 Business Opportunity

The SMRT Vocab fills a much-needed gap in the language learning software market by providing a renewed and fresh approach to one of the hardest parts of language learning. Vocabulary is an infinite endeavor among language learners of all kinds and levels. SMRT Vocab aims to help beginners and experienced, fluent learners who want to improve their language proficiency and reading levels. It is based on a body of linguistic research to fit into and bridge the proven methods of language learning while maintaining a humble focused goal. The principles of the application's design are; more progress, clear goals, less unneeded crazed features, and an efficiency which will drive learners to spread it to others. SMRT Vocab intends to utilize Zipf's Law and the Pareto Principle which appear in corpus linguistics to guide the SRS of the application. This will push learners to a level where CI is possible and achieve a Zone of Proximal Development for any target language. Zipf's law is a proven observation across all languages that describes the frequency and order of the most common words of a language (Zipf, 2013). Research has shown that word count and knowledge have significant impacts on learner comprehension levels. (See References Sections 1.3.3-1.3.6)

2.2 Problem Statement

The problem of	current language learning applications is that they all feature either complicated statistical tools or gamified applications which become inefficient and un-educational or overly monetized and bloated with useless features and
affects	language learners of all levels who desire to achieve comprehensible input and active use of practice and application
the impact of which is	that learners must begrudgingly use ineffective methods to acquire language vocabulary with little to no progress or time effectiveness
a successful solution would be	a user friendly, uncomplicated application which enhances the least enjoyable part of language acquisition with a clear goal and purpose to progress users to the next milestone in their proficiency levels which is lightweight, easily ported to various devices, and motivates learners through their own measurable and evidential.

2.3 Product Position Statement

For	language learners, students, and educational organizations to supplement practice of foreign languages
Who	need a tool to quickly achieve CI level vocabulary knowledge and to enhance reading levels
The SMRT Vocab application	is a language learning vocabulary training software
That	is designed to provide a path of least resistance to language proficiency and
Unlike	previous trendy applications or desktop software
Our product	cuts the fat out of the overall process of language learning by automating the vocabulary training aspect by using a newer SRS to automate learning words in sequence of frequency all contained in a small, undistracting program specifically designed to put all necessary tools within 3-5 clicks of the user picking up their device.

3. Stakeholder and User Descriptions

This section presents the profiles of the stakeholders and users involved in the project and the key problems they perceive to be addressed by the proposed solution.

3.1 Stakeholder Summary

Name	Description	Responsibilities
Academic Researchers	Academic personnel of linguistic areas who analyze and publish research to refer to language acquisition	Develops vocabulary databases based on the frequency of words in the language. Reviews and verifies vocabulary matchups for

	methods.	accuracy. It further develops and verifies methods and statistics because of observed application use.
Academic Leaders	Officials of academic institutions who look to provide resources to teachers, professors, and students.	Find and provide resources and tools to support their student's success.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Instructors	Teaches vocabulary, syntax, and grammar of a language.	Uses the application as a tool to facilitate linguistic understanding of vocabulary to reach the goal of language comprehension through yielding results.	Academic Leaders.
Students	Develops comprehension of a language to reach the target level of education.	Uses the application with guidance from their instructor to learn the vocabulary of a language based on the frequency of different words.	Academic Leaders.
Hobbyists	Learners with no mandatory obligation to pursue comprehension of an additional language to their first and do so in volition of their own interest.	Use the application productively to reach their goals.	NA

3.3 User Environment

Requirements of the working environment: The application will need internet working on a desktop PC or laptop. Ideally there will be a single device for each user. SMRT Vocab will be a desktop application with similar hardware requirements to Windows Operating Systems.

Time Commitment: The task cycle can be as short or as long as the users require based on their target reading level.

Persons in use: The application can either be used by a sole person, or as a collaborative tool used by many, such as a teacher using it on the projection of the classroom.

Platform Integrations: Integration is needed with the online vocabulary database which provides the word matchups for the application's walking window.

3.4 Summary of Key Stakeholder or User Needs

Key Problems	Stakeholders/Users	Reasons	Solution

Difficulty effectively learning target language vocabulary	Instructors, Students, and Hobbyists	Some techniques, methods, and approaches to acquisition of words are not easily understandable.	Synthesis and representation of research to defend theorized methods used by application developers for user enrichment and motivation into use of SRS.
Superfluous and misdirecting features in all applications available to users	Instructors, Students, and Hobbyists	Language coursework is weighed down with seemingly random vocabulary topics, modes, or games that do not effectively improve proficiency.	Focus on statistically supported words in the order of frequency as the primary measurement for language acquisition and progress. Ability to set, review, and change goals as desired.
Lack of tools to quickly verify and assess word count effectiveness on second language acquisition	Academic Researchers	Historical research is primarily observational in studies analyzing word count and frequency effects on language learning methods.	Application would provide a small effective and focused tool to test and measure the effects of supplementing various methods with frequency-based vocabulary.
Lack of tools to launch student comprehension aiding classroom lecture	Instructors, and Students	Traditional language education focuses on grammar but without a base of words to observe grammar this knowledge of words to apply it too grammar is not understood or retained.	Use of application as part of coursework will create a stronger pool of examples and knowledge to enhance instruction and acquisition of languages .
Significant cost of entry into language learning	Students and Hobbyists	Applications to supplement study are monetized for the inclusion of features, and academic realms take large amounts of time and tuition for education both of which are ineffective.	The applications minimalist design will provide a product at significantly lower cost to consumers as appropriate.

3.5 Alternatives and Competition

SMRT faces multiple challenges in the field of language learning: there are strong software competitors in the market of language learning such as Duolingo, Lingvist, and Anki Deck, many different courses and classes are available for different languages, and there are other free online sources such as YouTube and Quizlet. All of these are choices the customer can choose to use for their language learning journey instead of SMRT.

Famous for its appealing UI and trendiness, Duolingo is the top competitor on application stores. Its history follows the internet and mobile application revolution, and has strengths in having a low operation cost, has both free and paid options, the UI is attractive, and it has a substantial reputation. However, the main downside is that it is not effective in teaching the target language. Lingvist is another competitor whose application is easy to use- but it is both expensive and slow. On the other side of the perspective is Anki Deck, which is low cost – free with the exception of desktop versions - but has a confusing and complicated U.I.

There are also other resources that are free, easy to use, have new content uploaded frequently, and are easily accessible such as YouTube and Quizlet. The downside is that YouTube has unverified content and ads, and neither YouTube nor Quizlet has a structured regimen for language learning- it is self-led.

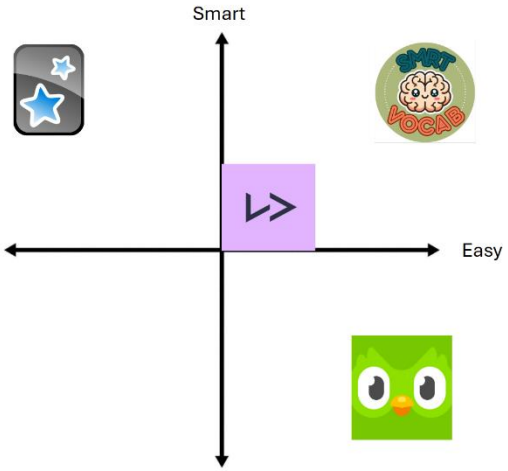


Image Source: Project Proposal Slides. SMRT Logo created by Malak Mahdy

4. Product Overview

This section provides a high-level view of the product capabilities, interfaces to other applications, and system configurations.

4.1 Product Perspective

SMRT Vocab is designed to be an easy to use and powerful language learning tool. By combining these two traits, this project aims to fill a gap in the language learning tool market that is dominated by applications that are either simple or powerful, but not both. It is designed to accomplish this through a simple interface and non-gamified learning techniques that are backed by language learning research. It is currently being developed as a desktop application, with plans to create a mobile application to increase its conveniency and make it a direct alternative to applications such as Duolingo. This product will be an independent and totally self-contained application.

4.2 Assumptions and Dependencies

4.2.1 System Requirements

Currently, this application is being developed for Windows PC using Python and its included interface package, tkinter. Python and tkinter are available on most Unix platforms, including macOS and Windows systems, so although Windows is the development environment, it will be possible to develop the application to be compatible with other operating systems if considerations for cross-platform compatibility are made.

4.2.2 Development Packages

The Python package tkinter is designed for desktop environments and is not suitable for mobile development on its own. Once mobile development begins, additional libraries will have to be included to support mobile environments. Mobile development will initially support Android OS, and support for iOS devices will require further considerations.

4.2.3 Network Requirements

The application will require an internet connection for initial installation and setup. The current version implementation will not require further networking or connections as profile information will be saved locally until further infrastructure is developed.

4.3 Cost and Pricing

The development team for this application consists of students, so minimizing cost is important. To accomplish this, the application will be developed using free and open-source software and libraries. However, man hours for all members, and implementation timeline estimates the cost of development to be approximately $5,000. This is based off of student research, studies, meetings, and personal development efforts as extracurricular activity. (7 members 4-7 hours per week for 10 weeks at $15/hour based on lower typical intern wages.)

The application will be free to download, making it consistent with other popular language learning applications. To financially support this application, there are several monetization strategies that may be employed.

- Freemium / Subscription based: A basic version of the application may be offered with either a one-time payment or monthly/annual subscription fee to access premium or advanced features, such as an ad-free experience or advanced statistics.
- Paid application: The desktop application may be free to use, and the mobile application will be a paid application.
- Advertisements: The free application may include ad spaces that can be sold to advertisers.
- Donations: The free application may include an option for users to donate money to the development team.

4.4 Licensing and Installation

The SMRT Vocab application should be properly licensed and comply with all regulations. The desktop application should be available for download through a release on the project's GitHub webpage. The mobile application should be available for download from the Google Play store and/or the Apple App Store. The application should be tested on different devices and platforms to ensure it works properly for all users.

5. Product Features

5.1 Feature 1 – Multiple Choice Flash Cards

Multiple choice flash cards: Users will be presented with a word from the language they choose to practice while being asked to select from a few words from a language they are familiar with.

5.2 Feature 2 – Typed-answer Flash Cards

Typed-answer flash cards: Users will be presented with a word from the language they choose to practice while being asked to type the word translated into a language they are familiar with.

5.3 Feature 3 – Pronunciation

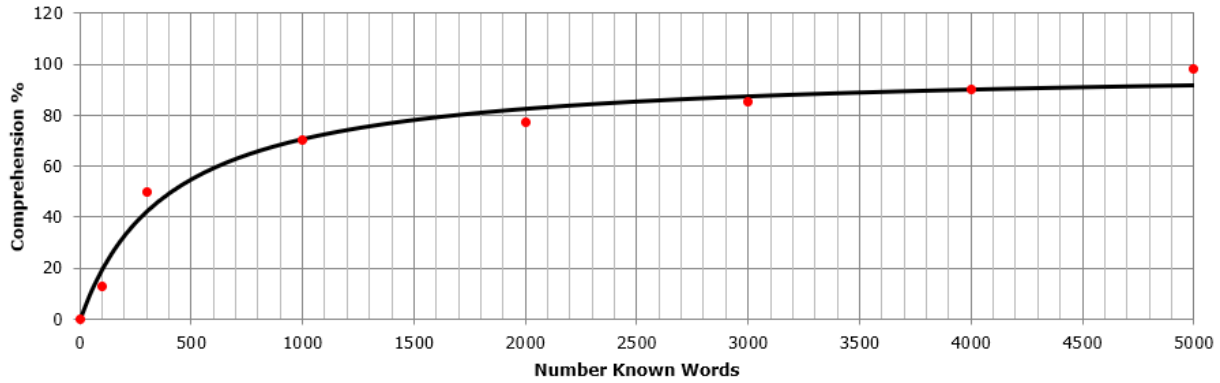
Pronunciation: When users are using one of the flash card features, they will have the option to listen to the spoken version of the word being learned or practiced.

5.4 Feature 4 – Count of Known Words

Count of known words: Users will be able to view how many of the 5,000 most common words they have learned.

5.5 Feature 5 – Percent Comprehension Graph

Percent comprehension graph: users will be able to view graphically what portion of the 5,000 most common words they have learned along with their estimated language comprehension percentage. Here is an example of our vision for this graph:



Graph Sources: Hazenberg, S., & Hulstijn, J. H. and Hwang

5.6 Feature 6 – Best words

Best words: Users will be able to view the five words that they have answered correctly most often.

5.7 Feature 7 – Worst words

Worst words: Users will be able to view the five words that they have answered correctly least often.

5.8 Feature 8 – New/Review Mode

Review mode: Users will have the option to choose between studying new words or reviewing words that have been marked as known.

5.9 Feature 9 – Mobile compatibility

Mobile compatibility: SMRT Vocab will be available on mobile devices.

5.10 Feature 10 – Translation Error Reporting

Translation error reporting: Users will have the option to report translation errors.

5.11 Feature 11 – User Registration and Login

User registration and login: Users will be able to register a username and password to save words they know, don't know, and are learning. Users will also be able to login with their username and password to retrieve their progress.

6. Constraints

There are various design constraints which are applicable to SMRT Vocab, including user information security, technical limitations, and language data.

6.1 Information Security

The handling of user information and data naturally places constraints on the design of the application. This includes securing the user's name, email, and login credentials such as username and password. Additionally, this may include payment information if in-app purchases are implemented in the future. All PII must be dealt with in compliance with strictest governmental policies and laws.

6.2 Technical and Design Limitations

There are different limitations from a technical perspective that apply to SMRT Vocab. Ensuring compatibility on different operating systems and devices increases accessibility for all its users. Furthermore, the user interface must remain clean and simple to enhance user experience.

6.2 Language Data

SMRT Vocab relies on the data of the top 5,000 words per language in order of word frequency. The languages offered will be dependent on the availability and accuracy of this information.

7. Quality Ranges

1. Performance: All features work with minimal delay and no failures in appearance or behavior.
2. Compatibility: Remain compatible with various operating systems and versions.
3. Security: Handle sensitive PII with the highest security measures.
4. Maintainability: Ability to easily add or edit features to increase optimal performance and easily updates.
5. Portability: Depending on implementations local profiles or web-based profiles synchronize correctly and can be carried over into mobile application versions.

8. Precedence and Priority

1. Primary learning algorithm implements flash cards correctly.
2. Primary Learning algorithm implements multiple choice cards correctly.
3. Primary learning algorithm progresses as intended and learners skip known words.
4. SRS aspects of algorithm function as intended for unknown words.
5. User registration and login
6. UI/UX is fluid, easy, and pleasing to use.
7. Creation of user profiles with usernames and passwords.
8. Learning progress and statistics are modelled effectively for users.
9. Review Algorithm behaves as defined.
10. Integration with cloud-based profile storage.
11. Mobile compatibility.
12. Ability for users to report errors in language data for developers to remedy.

Lang Gang

SMRT Vocab
Project Plan

Version 2.0
11/21/24

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
The business goals are about how to secure the team/company's place in the market. The product goals are about all the goals of the product. The quality goals list all the product quality attributes for the product.	YES
The deliverables list all the deliverables including all project artifacts (document, code, presentation, etc.).	YES
The project risks appropriately and in full sentences.	YES
The work package table is completed with all necessary fields.	YES

Final Control	Yes/No
The document is checked based on the “Team Project Artifacts Common Errors” Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Garrett Strader	Project Design Manager	Garrett Strader	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/24
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/24
Marcelo Barahona	Software Engineer	Marcelo Barahona	11/21/24
Voss Purkey	Project Manager	Voss Purkey	11/21/24

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Garrett Strader	Project Design Manager	Garrett Strader	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/24
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/24
Marcelo Barahona	Software Engineer	Marcelo Barahona	11/21/24
Voss Purkey	Project Manager	Voss Purkey	11/21/24

Revision History

Date	Version	Description	Author
09/20/24	1.0	The first iteration.	Voss Purkey
09/24/24	1.1	Added user registration and login work package to Project Work Breakdown Structure	Jacob Poland
10/20/24	1.2	Added table summary with hypothetical hourly rate to Budget section	Zachary Kao
10/21/24	1.3	Re-worded sections 1.1 & 1.2 Re-worded section 3.2 Removed developers from user stakeholder summary & section 3.4 Removed section 3.5	Voss Purkey
10/21/24	1.4	Added deliverables D6, D7, D8, and D9.	Ryleigh Sibley
10/22/24	1.5	Re-worded R1, R2, and R3 in section 10.	Ryleigh Sibley
11/11/24	1.6	Re-worded sections 1 & 2 expanding business goals. Reviewing of 3.4 and formatting and grammar corrections.	Voss Purkey
11/13/24	1.7	Fixed the signature fonts. Updated reference numbering. Removed a reference to security requirements in section 2.1 to reflect the SRS document. Revised sections 3, 7, and 8 so that they use full sentences.	Jacob Poland
11/21/24	2.0	Final check for submission	Malak Mahdy Marcelo Barahona

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Project Plan

1. Introduction

1.1 Purpose of the Document

The objective of this document is to outline the development plan for the SMRT Vocab application. It details the team's operations, organizational structure, and efforts to ensure the successful achievement of project goals. Updated versions of this document will serve as progress points throughout the product's lifecycle, providing a record of key decisions made during the development process. Further technical specifications and design details of the SMRT Vocab application will be available in the Project Design Document.

1.2 Scope of the Document

This Project Plan document outlines the goals and objectives of the SMRT Vocab application. It serves as a foundational resource for project planning, defining project objectives, and ensuring the proper implementation of the application. Additionally, it provides an overview of key stakeholders, users, and prioritizes product features within the Lang Gang team's operational framework. This document will serve as a guide for team members and project managers, directing efforts, and resource allocation. It will also facilitate the communication of any changes, ensuring all stakeholders are informed of development updates.

1.3 References

1.3.1 Lang Gang (2024). Internal Document: Project Proposal PowerPoint. Point of Contact for this document: Voss Purkey – Project Manager.

1.3.2 Lang Gang (2024). Internal Document: SMRT Vocab Project Vision Document. COSC3370_Lang Gang_Project_Vision_Document. Point of Contact for this document: Voss Purkey – Project Manager.

1.4 Definitions, Acronyms, and Abbreviations

See Appendix A.

2. Project Goals

2.1 Business Goals

There are various opportunities for business for a vocabulary-based language learning application. Therefore, the business goal related to SMRT Vocab is to increase the amount of registered, active users to a predicted number. In terms of profit, designated features that are attractive to the user will be provided with a subscription, creating a premium model for the business. Additionally, SMRT Vocab will create a strong presence in the market, making it the go-to application for language learning. Initial phases of the application launch will focus on testing, proving, and optimizing vocabulary acquisition. The potential exists for integration of the learning algorithm into various established organizations internationally for developing validity, and brand quality that other language learning pedagogy will want to include.

2.2 Product Goals

SMRT Vocab will stand out from its competitors through its simple user interface, making it accessible for learners of all levels. It will also provide an excellent supplement to in-class instruction, by allowing students to solidify what they learn in class and give them the tools needed to get ahead. SMRT Vocab is intended to easily provide the tools and aids needed, not hide or limit them behind games or rigid learning paths. Additionally, personalization based on the users' strengths and weaknesses will speed up the learning process. It also avoids gamification, to prevent the user from learning how to win the games instead of learning the language itself.

2.3 Quality Goals

SMRT Vocab maintains high-quality by concentrating on three domains: reliability, accuracy and modularity. SMRT Vocab strives to remain reliable and accessible in any circumstances to amend learning. It will retain the highest-quality for cross-platform accessibility, minuscule downtime, and quick response time regardless of the size

of the data set. Accuracy of the information and translations lay the primary foundation for the application; therefore, it is crucial to maintain the highest quality through thorough verifications.

3. Stakeholder and User Descriptions

This section presents the profiles of the stakeholders and users involved in the project and the key problems they perceive to be addressed by the proposed solution.

3.1 Stakeholder Summary

Name	Description	Responsibilities
Academic Researchers	Academic researchers are academic personnel of linguistic areas who analyze and publish research to refer to language acquisition methods.	They develop vocabulary databases based on the frequency of words in the language. They review and verify vocabulary matchups for accuracy. They further develop and verify methods and statistics because of observed application use.
Academic Leaders	Academic leaders are officials of academic institutions who look to provide resources to teachers, professors, and students.	They find and provide resources and tools to support their student's success.
Instructors	Instructors teach vocabulary, syntax, and grammar of a language.	They use the application as a tool to facilitate linguistic understanding of vocabulary to reach the goal of language comprehension.
Students	Students develop their comprehension of a language to reach the target level of education.	They use the application with guidance from their instructor to learn the vocabulary of a language based on the frequency of different words.
Hobbyists	Hobbyists are learners with no mandatory obligation to pursue comprehension of an additional language to their first and do so in volition of their own interest.	They use the application productively to reach their goals.

3.2 User Summary

Name	Description	Responsibilities	Stakeholder
Instructors	Teaches vocabulary, syntax, and grammar of a language.	Instructors use the application as a tool to facilitate linguistic understanding of vocabulary to reach the goal of language comprehension through yielding results.	Academic Leaders.
Students	Develops comprehension of a language to reach the target	Students use the application with guidance from their instructor to learn the vocabulary of a language based on the frequency of different	Academic Leaders.

	level of education.	words.	
Hobbyists	Learners with no mandatory obligation to pursue comprehension of an additional language to their first and do so in volition of their own interest.	Hobbyists use the application productively to reach their goals.	N/A

3.3 User Environment

Requirements of the working environment: Working internet and either a desktop or mobile device are required for each user.

Time Commitment: The task cycle can be as short or if the users require based on their target reading level.

Persons in use: The application can either be used by a sole person, or as a collaborative tool used by many, such as a teacher using it on the projection of the classroom.

Platform Integrations: Integration is needed with the online vocabulary database which provides the word matchups for the application's walking window.

3.4 Summary of Key Stakeholder or User Needs

Key Problems	Stakeholders/Users	Reasons	Solution
Learners have difficulty effectively learning target language vocabulary.	Instructors, Students, Hobbyists	There are not many easily understandable sources to show techniques, methods, and approaches to acquisition of words.	Synthesis and representation of research is used to defend theorized methods of application development for user enrichment and motivation into use of SRS.
Superfluous and misdirecting features are included in all applications available to users.	Instructors, Students, Hobbyists	Language coursework is weighed down with seemingly random vocabulary topics, modes, or games that do not effectively improve proficiency.	SMRT Vocab will focus on statistically supported words in the order of frequency as the primary measurement for language acquisition and progress. It will potentially include ability to set, review, and change goals as desired.
Lack of tools to quickly verify and assess word count effectiveness on	Academic Researchers	Historical research is primarily observational in studies analyzing word	Application would provide a small effective and focused tool to test

second language acquisition.		count and frequency effects on language learning methods.	and measure the effects of supplementing various methods with frequency-based vocabulary.
Lack of tools to launch student comprehension which will aid all realms of learning.	Instructors, students	Traditional language education focuses on grammar but without a base of words to observe grammar this knowledge of words to apply it too grammar is not understood or retained.	Use of application as part of coursework will create a stronger pool of examples and knowledge to enhance instruction and acquisition of languages .
Significant cost of entry into language learning.	Students, hobbyists	Applications to supplement study are monetized for the inclusion of features, and academic realms take large amounts of time and tuition for education both of which are ineffective.	The applications minimalist design will provide a product at significantly lower cost to consumers as appropriate.

4. Project Budget

The project is 12 weeks long and the development team has 7 members who can be expected to dedicate 3 hours per week towards the project, making the total effort budget for this project 252 man*hours. This effort budget will be spent on the following components:

- Documentation: There are 5 documents to be delivered upon completion of the project. If each team member budgets up to 1 hour toward the completion of each document and up to 1 hour to the revision of the documents for final delivery, the budget for creating documentation for the project is 42 man*hours.
- Planning: Each team member will be expected to dedicate some time to planning in the form of attending meetings. If each team member budgets 30 minutes to attending meetings each week, the budget for planning is 42 man*hours.
- Design: The team is expected to spend some time designing the systems for the project, particularly regarding the software requirements and design. The team will budget 30% of the remaining effort budget on design, or 50 man*hours.
- Development: The team is expected to spend some time developing the project, which includes coding and integrating components. The team will budget 50% of the remaining effort budget on development, or 84 man*hours.
- Testing: The team is expected to spend some time performing several tests on the project, including unit and integration tests. The team will budget the remaining 20% of the effort budget on testing, or 34 man*hours.

The following table summarizes the total cost for each component, based on a hypothetical hourly rate of $25 per team member:

Component	**Man*Hours**	**Hourly Rate ($)**	**Total Cost ($)**
Documentation	42	25	1050

Planning	42	25	1050
Design	50	25	1250
Development	84	25	2100
Testing	34	25	850
Total	**252**		**6300**

5. Project Milestones and Schedule

Project Start Date: 08/26/24

Project End Date: 11/25/24

Project Milestone	Date	Deliverable
Problem Analysis, Vision Creation, Project Planning	20 September 24	Product Vision Document Project Plan
Requirements Analysis	27 September 24	Software Requirements Specification (SRS)
Product Implementation Checkpoint 1	04 October 24	Software Code Iteration 1
Product Design	25 October 24	Software Design Document (SDD)
Product Test	01 November 24	Software Test Document (STD)
Project Implementation Checkpoint 2	08 November 24	Software Code Iteration 2
Final Product Deliverable	18 November 24	All Documentation and Code
Product Launch	19 November 24	Project Presentation

6. Project Communications

Project Communication Type	Responsible Team Member	Receiving Stakeholder	Frequency	Medium	Scheduled Time
Daily Project Communication	All Team Members	All Team Members	Anytime	Discord, E-mail, etc.	Anytime
Meetings	Project Manager / Team member	All Team Members	Twice a Week / as needed	Face-to-face / Online Zoom	Every Tuesday and Thursday after class. Virtual or in person meeting scheduled along with group availability
Project Progress Reports	Project Manager	All Team Members	Twice a week	E-mail, Discord	Every Tuesday and Thursday after class.
Project Effort Reports	All Team Members	All Team Members	As needed	Discord	Pushed into discord upon completion of

					function/assignment or when struggling with a problem.

7. Project Members and Roles

Project Role	Name	Responsibilities
Project Manager	Voss Purkey	Voss Purkey is responsible for creating the project plan, defining the project scope, allocating team members, planning resources, tracking milestones, managing issues, and ensuring successful project delivery.
Project Technical Manager	Ryleigh Sibley	Ryleigh Sibley is tasked with creating and maintaining the technical roadmap, solving technical problems, ensuring quality control, monitoring technical progress, and implementing process improvements.
Project Design Manager	Garrett Strader	Garrett Strader defines the design vision, leads the design direction, assures design quality, standardizes UI components, resolves design-related issues, and promotes continuous design improvements.
Project Test Manager	Zachary Kao	Zachary Kao develops the test strategy, oversees test execution, maintains testing standards, and ensures quality assurance and control throughout the project.
Software Configuration Manager	Jacob Poland	Jacob Poland manages the version control system, tracks changes, oversees change control and auditing, and drives continuous improvement within configuration management.
Software Requirement Manager	Malak Mahdy	Malak Mahdy documents requirements, prioritizes them, creates visual models, and ensures that all requirements align with the project scope.
Software Engineer	Marcelo Barahona	Marcelo Barahona provides general assistance with coding, programming, and supports all other roles within the team as needed to ensure optimal project outcomes.

8. Project Work Breakdown Structure (WBS)

Work Package ID	Work Package Name	Responsible Team Member	Estimated Effort (Man*Hour)	Definition
WP-1	Backend Module – Primary Learning Algorithm SUB WPs WILL BE ADDED HERE	Ryleigh Sibley, Voss Purkey, Zachary Kao. Jacob Poland, Malak Mahdy, Garrett Strader	70	This work package contains the development of the core algorithm for the SMRT Vocab application.
WP-2	Backend Module – API Development	Garrett Strader, Marcelo Barahona	20	This work package contains the work required to connect the core

				algorithm to the front end.
WP-3	User registration and login	Garret Strader and Jacob Poland	10	This work package contains the work required to create a functional user registration and login interface.
WP-4	User-Interface Module - Learning pages and placeholders	Garrett Strader	70	This work package contains the development of the multiple-choice and text-input flashcards features.
WP-5	Product Management	Voss Purkey	35	This work package contains the work associated with following up with team members, providing documentation, planning meetings, and ensuring timeline to launch, and clarifying customer requirements.
WP-6	UML Class Diagrams	Voss Purkey, Marcelo Barahona	3	This work package contains the development of UML diagram to describe interconnected functions and classes within the applications.
WP-7	Document Revision and Editing	All	100	This work package contains the creation and management of all deliverables.
WP-8	Configuration and Version Control	Jacob Poland	7	This work package contains the creation and management of git controls for SMRT Vocab's source code.

9. Project Deliverables

Deliverable Number	Deliverable Name
D-1	Project Vision
D-2	Project Plan
D-4	Software Requirements Specification
D-4	Software Design Document
D-5	Software Test Document
D-6	Team Project Presentation
D-7	SMRT Vocab Application Source Code
D-8	SMRT Vocab Application Download Instructions
D-9	SMRT Vocab Application Download File

10. Project Risks

Risk Number	Risk	Risk Reduction Approach
R-1	The user stands to lose progression due to application closure.	Progression saving will be implemented through data connected with a user's login.
R-2	Users could doubt the legitimacy of SMRT Vocabs teaching method compared to other competitors.	To dispel doubts, transparent information about the research and theories behind the application will be provided and made public.
R-3	There could be potential content errors such as misspelling or translation mistakes due to human error.	Quality content will be ensured through a thorough review process of vocabulary before being published.

11. Appendix A – Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Statistically Motivated Repetition Trainer.	SMRT	Name of project/program.
Comprehensible Input	CI	Language learning technique that uses a base knowledge and includes unknown words amended by context via images or other known words. This is close to methods such as the direct method or the natural method.
Spaced Repetition System	SRS	The name given to the system of smarter flashcard inspired learning which considers the frequency of views of a point of knowledge, the level of ease it is learned, and the number of times needed to prove a user knows it.
Personally Identifiable Information	PII	Refers to any data that can be used to identify an individual, such as names, addresses, Social Security numbers, or biometric records.
Zipf's Law	-	Statistical Law that reappears in various places. In linguistics this law describes the positive relation between a word's frequency and the sequential order of the following words.
Pareto Principle	-	Business principle described by Pareto as the 80/20 rule. It is associated with Ziph's law. An example would be that 80% of the effort in language learning accounts for 20% of the success or 20% of employees do 80% of the work.
Zone of Proximal Development	-	Pedagogical principles are referenced in various language learning methods. The principle states that people learn automatically via context and understanding when they understand around 98% of the content. This is associated with CI.
Application Programming Interface	API	The "interface" through which the backend and front-end modules will communicate IE. The front-end listeners trigger the retrieval of a stored word from the back-end, to be accessed and displayed to the user.
Learning Algorithm, Core algorithm	-	The base learning feature of the SMRT Vocab application which is the primary feature and based on the statistics of language learning
Unified Modeling Language	UML	A software development tool for reference, communication, and clarification. It defines the model of a software including the logic, functions, classes, and data.

Lang Gang

SMRT Vocab
Software Requirements Specifications (SRS)

Version 4.0
11/21/24

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the "Justify Text" format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states "The first version". Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
The requirements are written in full sentences.	YES
The requirements are atomic, clear, unambiguous, feasible, and testable.	YES
The requirements start with "The system shall…".	YES
The total number of "Frequency of Use" in the use cases add up to 100%.	YES
Each action in the use cases has a separate sequence number.	YES
The number of requirements is adequate to specify the details of all the features of the product.	YES
The requirements traceability matrix is complete.	YES
Each use case listed in the use case table is detailed with the use case template.	YES
All fields in the use cases are complete.	YES
In the use cases, the user actions start with "The user …" and the system actions start with "The system …".	YES
In the use cases, the steps in each alternative scenario include the same id structure such as 2.a., 3.a., 4.a., etc.	YES
All requirements have correct ID numbers.	YES

The use case names start with a verb followed by a noun.	YES
All use case actions have a sequence number, and all action sequence numbers are unique.	YES
The use case scenarios include an adequate number of actions to describe the scenario.	YES
The use case covers all the necessary feature-related scenarios.	YES

Final Control	**Yes/No**
The document is checked based on the "Team Project Artifacts Common Errors" Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/24
Marcelo Barahona	Project Risk Manager	Marcelo Barahona	11/21/24

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Garrett Strader	Project Design Manager	Garrett Strader	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Marcelo Barahona	Project Risk Manager	Marcelo Barahona	11/21/24
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/24
Voss Purkey	Project Manager	Voss Purkey	11/21/24
Malak Mahdy	Software Requirements Manager	Malak Mahdy	11/21/24

Revision History

Date	Version	Description	Author
09/27/24	1.0	Initial draft.	Voss Purkey Ryleigh Sibley Garrett Strader Jacob Poland Malak Mahdy Marcelo Barahona Zachary Kao
11/10/24	2.0	Elaborated on the introduction, scope of the document and use cases. Overall language refined, including spelling and punctuation.	Malak Mahdy
11/11/24	3.0	Updated requirements.	Garrett Strader
11/13/24	3.1	Added Req-Func-Sw-12 Added UC-10 Applied justified text format. Expanded sections 4 and 6. Updated traceability matrix. Verified all use case frequencies add to 100%.	Zachary Kao
11/15/24	3.2	Replaced references to “the application” with “the system.”	Jacob Poland
11/21/24	4.0	Final check for submission	Malak Mahdy Marcelo Barahona

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Software Requirements Specification (SRS)

1. Introduction

The requirements for the SMRT Vocab application, including its functionality, and design, are outlined in the Software Requirements Specifications (SRS) document. This document plays a vital role in guaranteeing the effectiveness and achievement of the system.

1.1 Purpose of the Document

The purpose of this document is to present the software requirements specifications for SMRT Vocab. It details the functional and non-functional requirements of the SMRT Vocab desktop application, encompassing its features, user interface, design elements, and technical specifications. This document is intended for the development team, project stakeholders, and any other parties who will benefit from a comprehensive understanding of SMRT Vocab's functionality and technical aspects.

1.2 Scope of the Document

The scope of the Software Requirements Specification (SRS) for the SMRT Vocab mobile application addresses every aspect of the app's functionality and design; including both technical and non-technical requirements.

The technical requirements for SMRT Vocab include specifications regarding the platform and operating system compatibility, the chosen programming language and development framework, and the hardware capabilities necessary for optimal performance.

In terms of non-technical requirements, SMRT Vocab's SRS covers user experience (UX) and user interface (UI) design expectations, adherence to branding standards, and any relevant legal or regulatory guidelines related to digital content, accessibility, and data privacy. Furthermore, the SRS outlines critical requirements related to application performance, scalability, and security to ensure a seamless and reliable user experience.

1.3 References

Reference Title/Name	Date	Version
Project Description Document	08/01/2024	2.0
Project Vision Document (PVD)	09/13/2024	1.0
Project Plan (PP)	09/20/2024	1.0
IEEE/ISO/IEC 29148-2018 - ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering	10/23/2018	2018
Web Content Accessibility Guidelines (WCAG)	11/11/2008	2.0

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Statistically Motivated Repetition Trainer.	SMRT	Name of project/program.

Comprehensible Input	CI	Language learning technique that uses a base knowledge and includes unknown words amended by context via images or other known words. This is close to methods such as the direct method or the natural method.
Spaced Repetition System	SRS	The name given to the system of smarter flashcard inspired learning which considers the frequency of views of a point of knowledge, the level of ease it is learned, and the number of times needed to prove a user knows it.
Personally Identifiable Information	PII	Refers to any data that can be used to identify an individual, such as names, addresses, Social Security numbers, or biometric records.
Zipf's Law	-	Statistical Law that reappears in various places. In linguistics this law describes the positive relation between a word's frequency and the sequential order of the following words.
Pareto Principle	-	Business principle described by Pareto as the 80/20 rule. It is associated with Ziph's law. An example would be that 80% of the effort in language learning accounts for 20% of the success or 20% of employees do 80% of the work.
Zone of Proximal Development	-	Pedagogical principles are referenced in various language learning methods. The principle states that people learn automatically via context and understanding when they understand around 98% of the content. This is associated with CI.
Application Programming Interface	API	The "interface" through which the backend and front-end modules will communicate IE. The front-end listeners trigger the retrieval of a stored word from the back-end, to be accessed and displayed to the user.
Learning Algorithm, Core algorithm	-	The base learning feature of the SMRT Vocab application which is the primary feature and based on the statistics of language learning.
Unified Modeling Language	UML	A software development tool for reference, communication, and clarification. It defines the model of a software including the logic, functions, classes, and data.

2. Product Scope

The product scope of the SMRT Vocab application is to help users expand their vocabulary of foreign languages through digital flashcards. th shall allow users to study words in various languages, track their progress, and reinforce learning with spaced repetition. Key features include studying from flashcards (multiple-choice answer selection or text-based response), support for learning multiple foreign languages, text-to-speech pronunciation, and customizable study settings. The primary focus is on enhancing vocabulary acquisition for language learners of all levels.

Out of scope for this product are advanced language learning features such as grammar lessons, conversational

practice, or full course material integration.

3. Product Users

3.1 Users

Product User Type	User Characteristics
Instructor User	• Age: Any age; typically, professionals in educational or instructional roles. • Education level: Must have at least an elementary understanding of the concept of 'reading'. Expected to possess educational fluency in the student's target language to facilitate teaching. • Technology level: Ranges from beginner to skilled; some may have extensive digital teaching experience, while others are adapting to new tools. • Role: Uses the system as a teaching tool, either working with students to practice vocabulary in a collaborative manner or assigning specific statistical goals to be achieved by students. • Other characteristics: May include strong communication skills, adaptability to new educational technologies, and a commitment to student progress.
Student User	• Age: Any age; from young children to adults pursuing new language skills. • Education level: Must have at least an elementary education and be familiar with the concept of 'reading'. • Technology level: Varies from beginner (learning digital tools) to skilled (digital natives familiar with various apps). • Role: Utilizes the platform to acquire and reinforce vocabulary in their target language. • Other characteristics: May exhibit a willingness to learn, dedication to achieving language goals, and the ability to adapt to new learning methods.
Hobbyist User	• Age: Varies widely; can range from children to adults pursuing new language skills, though typically older in age. • Education level: At least an elementary education in reading; often motivated learners with various levels of self-study experience. • Technology level: Ranges from beginner (learning digital tools) to skilled (digital natives familiar with various apps). • Role: Engages with the system independently to meet personal language-learning objectives. • Other characteristics: Often characterized by intrinsic motivation, passion for self-improvement, and the ability to work independently within a structured learning environment.

Academic Researcher Users	• Age: 18 years or older, typically adult professionals or students with a research focus. • Education level: Must have at least a high school education and often hold a bachelor's degree or higher, with proficiency in verifying languages they specialize in. • Technology level: Ranges from intermediate (comfortable with using and analyzing app-based data) to skilled (adept at software, statistics, or educational research tools). • Role: Responsible for reviewing, verifying, and validating vocabulary matchups in the system, observing usage patterns, and developing statistical methods and reporting mechanisms to benefit all user types. • Other characteristics: May include advanced analytical skills, critical thinking, strong attention to detail, and a focus on continuous improvement through research.

4. Use Cases

Use Case ID	Use Case Name	Brief Description
UC-1	Registration	This use case describes the user registration process, where users create an account by providing a username and password for unique identification. Upon successful registration, the system initializes word lists and settings.
UC-2	Login	This use case describes how users enter their registered credentials to access the platform. If verified, users are redirected to their personalized dashboard; otherwise, they are prompted to retry or register their account if they are a new user.
UC-3	Menu Navigation	This use case allows users to navigate through the core menu options including multiple-choice flashcards, text-input flashcards, statistics, and settings. Selecting the button will immediately take the user to its respective use case.
UC-4	Multiple-Choice Flashcards	This use case enables users to engage with multiple-choice flashcards by selecting the correct answer from several options for a presented word or phrase. Immediate feedback, including correctness and explanations, is provided.
UC-5	Text-Input Flashcards	This use case describes how users translate presented words or phrases by typing their answers, which the system checks for accuracy. Immediate feedback, including correctness and explanations, is provided.
UC-6	View Statistics	This use case describes how users can view detailed statistics showing the number of known words, their top-performing vocabulary, and progress toward overall language mastery. This feature helps track learning milestones and identify areas needing improvement.
UC-7	Settings Editing	This use case describes how users can customize the criteria for marking words as "known," including setting specific thresholds and deltas. Additionally, users can modify the spaced repetition queue length and adjust the walking window

		size, select a new language to learn, and choose whether to practice translating from the foreign language to English or vice versa.
UC-8	Text to Speech	This use case describes how users can hear a word's pronunciation through text to speech. The user can also hear a word's pronunciation automatically when loading the page when a setting is set.
UC-9	Mark Known Words	This use case describes how users manually mark words as "known" through clicking a button, to make study sessions more effective. This feature adjusts future content to focus on unfamiliar vocabulary.
UC-10	Review Mode	This use case describes how users can use the review feature to study words that are already marked as known. Words are reviewed by using both flashcard study methods at random. Immediate feedback, including correctness and explanations, is provided.

5. Functional Requirements

This section provides a list of software and hardware requirements.

5.1 Software Requirements

Requirement ID	Requirement Priority (High/Medium/Low)	Requirement
Req-Func-Sw-1	High	The system shall allow users to create an account and log in using their credentials, if valid.
Req-Func-Sw-2	High	The system shall allow users to match a definition to a word in a flashcard format using mouse clicks.
Req-Func-Sw-3	Medium	The system shall provide a text input field during the study session that allows the user to manually enter a word's definition.
Req-Func-Sw-4	High	The system shall provide the proper feedback when the user provides the correct or incorrect answers.
Req-Func-Sw-5	High	The system shall provide a button for each menu (Multiple Choice Flashcards, Text-Input Flashcards, Statistics, Settings, Back, and Exit) that will navigate to the designated menu.
Req-Func-Sw-6	Low	The system shall provide a button that will play a text-to-speech pronunciation of a flashcard word when pressed.
Req-Func-Sw-7	Medium	The system shall provide the user with a count of the words they have learned.
Req-Func-Sw-8	Medium	The system shall provide the user with a graphical representation of their estimated language comprehension percentage.
Req-Func-Sw-9	Low	The system shall allow users to view the word they have answered correctly the most, as well as the least.
Req-Func-Sw-10	Medium	The system shall give users an option to switch between studying new words and reviewing words they have learned.
Req-Func-Sw-11	Low	The system shall provide users with options for various changes through the settings menu.
Req-Func-Sw-12	High	The system shall save the user's learning progress between use sessions.

5.2 Hardware Requirements

Requirement ID	Requirement Priority (High/Medium/Low)	Requirement
Req-Func-Hw-1	High	The system shall operate with a computing device with at least 8 GB RAM for optimal performance.

Req-Func-Hw-2	Medium	The system should have at least 1GB of available storage space for data storage and system files with room for expansion into other languages with future updates. The system will save user data and audio files to local storage, so the system must have adequate storage available.
Req-Func-Hw-3	Low	The system should have a display monitor with a minimum resolution of 1280x1024 pixels for clear and comfortable viewing.
Req-Func-Hw-4	High	The system should support multi-core processing (at least quad core) to handle multiple simultaneous processes efficiently, enhancing overall system responsiveness and speed.

6. Quality Requirements

Requirement ID	**Requirement Priority** (High/Medium/Low)	**Requirement**
Req-Qual-Perf-1	High	The system shall ensure that all application pages load within 3 seconds to maintain user engagement.
Req-Qual-Perf-2	High	The system shall process user responses and provide feedback and testable words within 0.5 seconds to enhance the learning experience.
Req-Qual-Use-1	High	The system shall comply with accessibility standards to ensure it is usable by learners with disabilities. Color-blindness is accounted for through the proper labelling of the user interface. Disabilities and reading impairment is accounted for through the availability of text-to-speech options.
Req-Qual-Use-2	Medium	The system shall provide intuitive navigation and a user-friendly interface to enhance the user experience. Navigation buttons should be properly labelled and employ the use of iconography to assist in understanding their use.
Req-Qual-Scal-1	Medium	The system shall support the addition of new languages and courses without significant changes to the existing infrastructure.

7. Appendix A – Requirements Traceability

Requirement ID	Related Requirements	Test Case ID
Req-Func-Sw-1	Req-Qual-Sec-1	TC-1
Req-Func-Sw-2	Req-Func-Sw-4 Req-Func-Sw-6 Req-Qual-Perf-1 Req-Qual-Perf-2	TC-3 TC-5
Req-Func-Sw-3	Req-Func-Sw-4 Req-Func-Sw-6 Req-Qual-Perf-1 Req-Qual-Perf-2	TC-4 TC-5
Req-Func-Sw-4	Req-Func-Sw-2 Req-Func-Sw-3	TC-3 TC-4 TC-5
Req-Func-Sw-5	Req-Qual-Use-1 Req-Qual-Use-2 Req-Func-Sw-10	TC-2 TC-3 TC-4 TC-5
Req-Func-Sw-6	Req-Qual-Use-1	TC-3 TC-4 TC-5
Req-Func-Sw-7		TC-6
Req-Func-Sw-8		TC-6
Req-Func-Sw-9		TC-6
Req-Func-Sw-10	Req-Func-Sw-11	TC-5 TC-6
Req-Func-Sw-11	Req-Qual-Use-1 Req-Func-Sw-10	TC-6
Req-Func-Sw-12	Req-Func-Sw-2 Req-Func-Sw-3 Req-Func-Sw-7 Req-Func-Sw-8 Req-Func-Sw-9 Req-Func-Sw-10	TC-3 TC-4 TC-5 TC-6

Req-Func-Hw-1		TC-2
Req-Func-Hw-2		TC-2
Req-Func-Hw-3		TC-2
Req-Func-Hw-4		TC-2
Req-Qual-Perf-1		TC-2
Req-Qual-Perf-2		TC-3 TC-4 TC-5
Req-Qual-Use-1		TC-2 TC-3 TC-4 TC-5
Req-Qual-Use-2		TC-2
Req-Qual-Scal-1		TC-6

8. Appendix B – Use Case 1 – Registration

Use Case ID	UC-1
Use Case Name	Registration
Brief Description	This use case describes the user registration process. The user will provide a username and password. The system will register the user and initialize their word lists.
Frequency of Use	Low - 5%
Priority	High
Current Version	1.0
Date of First Version	09/26/2024
Date of Last Version	09/26/2024
Created By	Jacob Poland
Last Update By	Jacob Poland
Approved By	Ryleigh Sibley
Assumptions	Users are capable and willing to save their credentials elsewhere. Each user uses SMRT Vocab on a single device. Users know how to use a mouse. The user may attempt to inject code into one or more text input fields.
Primary Actor	User
Secondary Actor/s	None

Preconditions	The user has SMRT Vocab downloaded. The Spanish table template is available. The system is experiencing normal working conditions.	
Postconditions	The user is registered in the system. The system displays the main menu.	
Trigger	The user clicks on “Register your account” in the welcome screen.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user launches SMRT Vocab.	2. The system displays the welcome screen.
	3. The user enters a username, password, and email.	
	4. The user selects “Register your account.”	5. The system displays a confirmation of successful registration.
	6. The user enters the user’s information.	
	7. The user submits the registration form.	8. The system validates the user’s information. 9. The system adds the user’s credentials to the registration table. 10. The system creates the table to save the user’s learning progress. 11. The system displays the main menu.
Alternate Scenarios	**User Actions**	**System Actions**
	6a. The user submits a taken username.	7a. The system notifies the user that the username is taken.
	8a. The user attempts to register again.	
	6b. The user submits a username that contains invalid characters.	7b. The system notifies the user that the username contains invalid characters.
	8b. The user attempts to register again.	

	6.c. The user submits a password that contains invalid characters. 8.c. The user attempts to register again.	7c. The system notifies the user that their password contains invalid characters.
Additional Notes (Constraints, etc.)	There will be no way for users to recover their accounts if their username or password is lost or forgotten. Additionally, learning progress will only be saved on the device the user registers on.	

Use Case 2 – Log In

Use Case ID	UC-2	
Use Case Name	Log In	
Brief Description	This use case describes the user login process. The user will provide their username and password. The system will verify the user's credentials.	
Frequency of Use	Medium - 15%	
Priority	High	
Current Version	1.0	
Date of First Version	09/26/2024	
Date of Current Version	09/26/2024	
Created By	Jacob Poland	
Last Update By	Jacob Poland	
Approved By	Ryleigh Sibley	
Assumptions	The user knows how to use a mouse and keyboard. The user has SMRT Vocab downloaded. The user may attempt to inject code into one or more text input fields. Each user uses SMRT Vocab on a single device.	
Primary Actor	User	
Secondary Actor/s	None	
Preconditions	The user is registered with the system. The system is experiencing normal working conditions.	
Postconditions	The user has SMRT Vocab downloaded.	
Trigger	The user starts SMRT Vocab and clicks on the log in button.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user starts SMRT Vocab.	2. The system displays the welcome screen that has two buttons: one button to 'Login' and the other to 'Sign Up/Register.'

	3. The user clicks the 'Login' button. 5. The user enters the user's username and password. The user presses enter or clicks 'Login' to submit their credentials.	4. The system displays the login screen that prompts the user for username and password. 6. The system verifies the user's credentials. 7. The system saves the username to a variable for future data retrieval. 8. The system displays the main menu.
Alternate Scenarios	**User Actions**	**System Actions**
	5a. The user submits incorrect credentials. 8a. The user reattempts to login or goes back to the welcome page to register. 5b. The user attempts to inject code through the username or password fields.	6a. The system verifies the provided credentials. 7a. The system notifies the user that the username or password in incorrect and that new users must register first. 6b. The system verifies the provided credentials. 7a. The system notifies the user that the username or password in incorrect and that new users must register first.
Additional Notes (Constraints, etc.)	There will be no way for users to recover their accounts if their username or password is lost or forgotten. Additionally, learning progress will only be saved on the device the user registers on. These notes are the same as UC-1.	

Use Case 3 – Menu Navigation

Use Case ID	UC-3
Use Case Name	Menu Navigation
Brief Description	This use case describes user navigation of menu options multiple choice, user entry, back, exit, and settings.
Frequency of Use	Medium - 15%
Priority	Medium
Current Version	1.0
Date of First Version	09/26/2024

Date of Current Version	09/26/2024	
Created By	Ryleigh Sibley	
Last Update By	Ryleigh Sibley	
Approved By	Zachary Kao	
Assumptions	The user knows how to use a keyboard and mouse.	
Primary Actor	User.	
Secondary Actor/s	None.	
Preconditions	The user is registered and logged into the system.	
Postconditions	The user navigates to a different feature of the system.	
Trigger	User Actions	
Main Success Scenario	1. The user clicks one of the following buttons: Statistics, Settings, Back, or Log Out.	2. The system displays the corresponding feature.
Alternate Scenarios	User Actions	System Actions
	1a. The user clicks the “Learning Mode” button. 1b. The user clicks the “Review Mode” button.	2a. The system saves the user’s choice to learn new words and displays the following buttons: “Multiple-Choice Flashcards” and “Text-Input Flashcards.” 2b. The system saves the user’s choice to review known words and displays the following buttons: “Multiple-Choice Flashcards” and “Text-Input Flashcards.”
Additional Notes (Constraints, etc.)	None	

Use Case 4 – Multiple-Choice Flashcards

Use Case ID	UC-4
Use Case Name	Multiple-Choice Flashcards
Brief Description	This use case describes user selection of multiple-choice flashcards and displaying if user selection was correct
Frequency of Use	High - 20%
Priority	High

Current Version	1.0	
Date of First Version	09/27/24	
Date of Current Version	09/27/24	
Created By	Garrett Strader	
Last Update By	Garrett Strader	
Approved By	Jacob Poland	
Assumptions	The user knows how to use a mouse. The system is working under normal conditions.	
Primary Actor	User	
Secondary Actor/s	None	
Preconditions	The user is on the Multiple-Choice Flashcard screen. The user has selected either Review Mode or Learning Mode.	
Postconditions	User selection is recorded as correct or incorrect. The count of Correct or Incorrect selections is updated in the user's word database. The user's known words are updated as needed.	
Trigger	The user clicks the "Multiple Choice Flashcards" button on main menu.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user clicks the "Multiple Choice Flashcards" button on the main menu.	
		2. The system randomly selects a flashcard word from the appropriate list based on the user's desired mode (review or learning).
		3. The system retrieves the correct translation of the flashcard word.
		4. The system retrieves three other words and their corresponding translations from the appropriate word list.
		5. The system displays the flashcard word and the back button.
		6. The system displays the correct translation as well as translations of three other words as buttons.
	7. The user clicks to select one of the four displayed translations.	
		8. The system checks the user's choice for correctness.

		9. The system increments the user's count of correct or incorrect attempts for the flashcard word. 10. The system marks the flashcard word as known in the user word table if the user is in learning mode and their count of correct attempts meets the set threshold. 11. The system displays 'correct' or 'incorrect' to the user. 12. The system displays the correct translation if the user's selection was incorrect. 14. The system loops back to step 1 of this main success scenario.
	13. The user clicks the "Continue" button.	
Alternate Scenarios	**User Actions**	**System Actions**
	7.a User selected back button. 13b. The user clicks the "Back" button.	8.a. The system will display the main menu. 14b. The system will display the main menu.
Additional Notes (Constraints, etc.)	None	

Use Case 5 – Text-Input Flashcards

Use Case ID	UC-5
Use Case Name	Text-Input Flashcards
Brief Description	This use case describes a flash card feature where users type a given word's translation, and the system verifies its correctness.
Frequency of Use	High - 20%
Priority	High
Current Version	1.0
Date of First Version	09/27/2024
Date of Current Version	09/27/2024
Created By	Jacob Poland
Last Update By	Jacob Poland
Approved By	Ryleigh Sibley
Assumptions	The user knows how to use a keyboard and mouse.

	The user may attempt to inject code into the text input field.	
Primary Actor	User	
Secondary Actor/s	None	
Preconditions	The user is registered with the system. The user is logged into the system. The database is free of data inconsistencies. The system is in normal working condition. The user has SMRT Vocab running. The user has selected either learning mode or review mode. The user has selected the “Text-Input Flashcards” button.	
Postconditions	The system displays if the user translation is correct or incorrect. The system displays the correct translation. The word count of correct or incorrect attempts to translate the given flashcard word is updated. The user’s list of known words is updated as necessary.	
Trigger	The user clicks on the ‘Text-Input Flashcard's’ button.	
Main Success Scenario	**User Actions**	**System Actions**
		1. The system randomly selects a flashcard word from the appropriate word list based on the user’s desired mode (review or learning). 2. The system retrieves the correct translation of the selected word. 3. The system displays the flashcard word and a text-input field.
	4. The user types their known language translation of the flashcard word. 5. The user clicks ‘Submit’, or presses Enter.	
		6. The system validates the input to protect against code injection. 7. The system compares the user-provided translation with the correct translation. The comparison is case insensitive. 8. The system increments the count of correct or incorrect translation attempts for the current word.

	12. The user clicks the 'Continue' button.	9. The system marks the flashcard word as known if the user is in learning mode and the count of correct translation attempts reaches the set threshold. 10. The system displays the correctness of the user's translation. 11. The system displays the correct translation if the user's translation was incorrect. 13. The system loops back to step 1 of this main success scenario to display the next flashcard.
Alternate Scenarios	**User Actions**	**System Actions**
	4a. The user clicks the "Back" button. 12a. The user clicks the 'Back' button	5a. The system displays the main menu. 13a. The system displays the main menu.
Additional Notes (Constraints, etc.)	The capitalization of user input will be normalized before checking it for correctness. Special characters in Latin script languages will not be mandatory.	

Use Case 6 – View Statistics

Use Case ID	UC-6
Use Case Name	View Statistics
Brief Description	This use case describes how a user views their statistics, including number of known words, their best-known words, and progress towards language comprehension.
Frequency of Use	Low – 5%
Priority	Medium
Current Version	1.0
Date of First Version	9/26/2024
Date of Current Version	9/26/2024
Created By	Zachary Kao
Last Update By	Zachary Kao
Approved By	Jacob Poland
Assumptions	The user knows how to use a mouse.
Primary Actor	User
Secondary Actor/s	None
Preconditions	The user is registered and logged in.

	The user is on the main menu screen. The system's database is in normal working condition. The user has data stored with the system.	
Postconditions	The system displays the user's statistics, including best and worst known words, number of known words, and a graphic representation of their progress towards language comprehension.	
Trigger	The user clicks the "View Statistics" button on the main menu.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user clicks the "Statistics" button.	2. The system navigates to the "View Statistics" screen. 3. The system fetches the user's data to calculate statistics. 4. The system calculates the user's 5 best-known words. 5. The system calculates the user's 5 worst-known words. 6. The system calculates the user's number of known words. 7. The system calculates the user's progress towards language comprehension based off the number of known words. 8. The system displays the user's statistics to the screen.
Alternate Scenarios	**User Actions**	**System Actions**
	None	None
Additional Notes (Constraints, etc.)	Progress towards language comprehension is based on the Pareto Principle and the Zone of Proximal Development. Percentages will follow a scale-based known word count as supported by most up to date research.	

Use Case 7 – Settings Editing

Use Case ID	UC-7
Use Case Name	Settings Editing
Brief Description	This use case describes how the user can adjust the standards of the 'known' status of a word.
Frequency of Use	Low - 5%
Priority	Low
Current Version	1.0
Date of First Version	09/27/2024

Date of Current Version	09/27/2024	
Created By	Ryleigh Sibley	
Last Update By	Ryleigh Sibley	
Approved By	Zachary Kao	
Assumptions	The user knows how to use a mouse.	
Primary Actor	User.	
Secondary Actor/s	None.	
Preconditions	The system is working under normal conditions. The user has successfully registered and logged in. The main menu is displayed.	
Postconditions	The system updates how many successful attempts trigger the condition for a word to have the 'known' state.	
Trigger	User clicks on the 'Settings' option in the main menu.	
Main Success Scenario	**User Actions**	**System Actions**
	1.The user selects the 'Settings' option in the main menu. 5.The user enters the number is successful attempts required for a word to gain the 'known' status.	2.The system receives the user's selection input of the main menu. 3.The system navigates to the 'Settings' window. 4.The system displays the 'Settings' menu to the user. 6. The system retrieves the input from the user. 7. The system updates the previously set condition for the 'known' status.
Alternate Scenarios	**User Actions**	**System Actions**
	None.	None.
Additional Notes (Constraints, etc.)	None	

Use Case 8 – Text to Speech

Use Case ID	UC-8
Use Case Name	Text to Speech
Brief Description	This use case describes how users can hear a word's pronunciation through text to

	speech.	
Frequency of Use	Low - 5%	
Priority	Low	
Current Version	2.0	
Date of First Version	09/24/2024	
Date of Current Version	11/13/2024	
Created By	Zachary Kao	
Last Update By	Zachary Kao	
Approved By	Garrett Strader	
Assumptions	The user is studying a word in either flashcard or text-input mode.	
Primary Actor	User	
Secondary Actor/s	None.	
Preconditions	The user is studying a word.	
Postconditions	The pronunciation of the word is played to the user.	
Trigger	The user presses the pronunciation button.	
Main Success Scenario	**User Actions**	**System Actions**
	1. The user presses the pronunciation button.	2. The system requests the pronunciation of the displayed word from the text-to-speech library. 3. The system checks if the desired pronunciation audio file already exists and finds no existing audio file. 4. The system generates a new pronunciation audio file and saves it to local storage. 5. The system plays the pronunciation of the word.
Alternate Scenarios	User Actions	System Actions
		3.a. The system checks if the desired pronunciation audio file already exists and finds an existing audio file. 4.b. The system plays the pronunciation of the word.
Additional Notes (Constraints, etc.)	The system should be able to handle the pronunciation of non-English words.	

Use Case 9 – Mark Known Words

Use Case ID	UC-9	
Use Case Name	Mark Known Words	
Brief Description	This use case describes how users can mark a word as “known” to better tailor their learning.	
Frequency of Use	Low - 5%	
Priority	Low	
Current Version	1.0	
Date of First Version	09/27/2024	
Date of Current Version	09/27/2024	
Created By	Malak Mahdy	
Last Update By	Malak Mahdy	
Approved By	Jacob Poland	
Assumptions	The user is accurately indicating what they know. The user knows how to use a mouse and keyboard.	
Primary Actor	User	
Secondary Actor/s	None.	
Preconditions	The user has just correctly translated a vocabulary flashcard.	
Postconditions	The word will no longer appear in the “still learning” flashcards.	
Trigger	The user clicks on the "Already Know” button.	
Main Success Scenario	User Actions	System Actions
	1. The user presses the “I know this word” button.	2. The system transfers the word from the “learning” words list to the “known” words list. 3. The system adds a new word to the “learning” words list.
Alternate Scenarios	User Actions	System Actions
	None.	None.
Additional Notes (Constraints, etc.)	None.	

Use Case 10 – Review Mode

Use Case ID	UC-10
Use Case Name	Review Mode

Brief Description	This use case describes how users can use the review feature to study words that are already marked as known. Words are reviewed by using both flashcard study methods at random. Immediate feedback, including correctness and explanations, is provided.	
Frequency of Use	Low - 5%	
Priority	Medium	
Current Version	1.0	
Date of First Version	11/12/2024	
Date of Current Version	11/12/2024	
Created By	Zachary Kao	
Last Update By	Zachary Kao	
Approved By	Jacob Poland	
Assumptions	The user knows how to use a mouse and keyboard.	
Primary Actor	User	
Secondary Actor/s	None.	
Preconditions	The user is registered with the system. The user is logged into the system. The database is free of data inconsistencies. The system is in normal working condition. The user has SMRT Vocab running. The user already has data with words marked as known.	
Postconditions	The system displays if the user translation is correct or incorrect. The system displays the correct translation. The word count of correct or incorrect attempts to translate the given flashcard word is updated.	
Trigger	The user clicks on the "Review Mode" button.	
Main Success Scenario	User Actions	System Actions
	1. The user presses the Review Mode button. 4. The user presses an answer.	2. The system displays the review mode page. 3. The system displays a review flashcard of a known word in the multiple-choice format. 5. The system checks the user's choice for correctness.

	10. The user presses OK or waits for the success message to disappear.	6. The system increments the user's count of correct or incorrect attempts for the flashcard word. 7. The system marks the flashcard word as known in the user word table if the user is in learning mode and their count of correct attempts meets the set threshold. 8. The system displays 'correct' or 'incorrect' to the user. 9. The system displays the correct translation if the user's selection was incorrect. 11. The system loops back to step 3 or 3.b to display the next flashcard.
Alternate Scenarios	User Actions	System Actions
	1.a. The user presses the Review Mode button when they have no words known. 4.b. The user types their known language translation of the flashcard word. 5.b. The user clicks 'Submit', or presses Enter.	2.a. The system displays a message stating that the user has no words to review. 3.b. The system displays a review flashcard of a known word in the text-input format. 6.b. The system validates the input to protect against code injection. 7.b. The system compares the user-provided translation with the correct translation. The comparison is case insensitive. 8.b. The system increments the count of correct or incorrect translation attempts for the current word. 9.b. The system marks the flashcard word as known if the user is in learning mode and the count of correct translation attempts reaches the set threshold.

	12.b. The user presses OK or waits for the success message to disappear.	10.b. The system displays the correctness of the user's translation. 11.b. The system displays the correct translation if the user's translation was incorrect. 13.b. The system loops back to step 3 or 3.b. to display the next flashcard.
Additional Notes (Constraints, etc.)	The display of either multiple-choice or text-input flashcards in review mode is random.	

Lang Gang

SMRT Vocab
Software Design Document (SDD)

Version 3.0
11/21/24

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
The design decisions, design constraints, and the related reasonings are written in full sentences.	YES
The system designs are provided in such detail to guide the remaining development process.	YES
The user interface designs are provided in such detail to guide the remaining development process.	YES
The modules are described with adequate descriptions.	YES

Final Control	Yes/No
The document is checked based on the “Team Project Artifacts Common Errors” Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
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Revision History

Date	Version	Description	Author
10/25/24	1.0	Initial draft.	Malak Mahdy
11/10/24	2.0	Updated UML diagrams. Added Settings module. Updated modules to reflect current implementation. Updated product scope. Improved document formatting.	Zachary Kao
11/13/24	2.1	Added ReviewGUI and AboutGUI to diagrams and tables as necessary. Re-organized UML Class diagram for readability. Added TestToSpeech file to SDD. Implemented general corrections and formatting.	Voss Purkey
11/18/24	2.2	Updated UI screenshots in section 8.	Jacob Poland
11/21/24	3.0	Final check for submission	Malak Mahdy Marcelo Barahona

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Software Design (SDD)

1. Introduction

1.1 Purpose of the Document

The purpose of this document is to document the software design for the SMRT Vocab application project. It provides a well detailed definition of the system under development.

1.2 Scope of the Document

This document will cover the design decisions and constraints of the SMRT Vocab project as well as the proposed architecture and user interface designs. It provides tables and diagrams explaining software logic, activities, and dataflow.

1.3 References

Reference Title/Name	Date	Version
Project Description Document	08/15/2024	2.0
Project Vision Document (PVD)	11/18/2024	3.0
Project Plan (PP)	11/18/2024	2.0
Project Software Requirements Specification (SRS)	11/18/2024	2.0
Project Software Test Description (STD)	11/20/2024	1.5
IEEE/ISO/IEC 29148-2018 - ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering	10/23/2018	2018
Web Content Accessibility Guidelines (WCAG)	11/11/2008	2.0

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Statistically Motivated Repetition Trainer.	SMRT	Name of project/program.
Comprehensible Input	CI	Language learning technique that uses a base knowledge and includes unknown words amended by context via images or other known words. This is close to methods such as the direct method or the natural method.
Spaced Repetition System	SRS	The name given to the system of smarter flashcard inspired learning which considers the frequency of views of a point of knowledge, the level of ease it is learned, and the number of times needed to prove a user knows it.
Personally Identifiable Information	PII	Refers to any data that can be used to identify an individual, such as names, addresses, Social Security numbers, or biometric records.

Zipf's Law	-	Statistical Law that reappears in various places. In linguistics this law describes the positive relation between a word's frequency and the sequential order of the following words.
Pareto Principle	-	Business principle described by Pareto as the 80/20 rule. It is associated with Ziph's law. An example would be that 80% of the effort in language learning accounts for 20% of the success or 20% of employees do 80% of the work.
Zone of Proximal Development	-	Pedagogical principles are referenced in various language learning methods. The principle states that people learn automatically via context and understanding when they understand around 98% of the content. This is associated with CI.
Application Programming Interface	API	The "interface" through which the backend and front-end modules will communicate IE. The front-end listeners trigger the retrieval of a stored word from the back-end, to be accessed and displayed to the user.
Learning Algorithm, Core algorithm	-	The base learning feature of the SMRT Vocab application which is the primary feature and based on the statistics of language learning
Unified Modeling Language	UML	A software development tool for reference, communication, and clarification. It defines the model of a software including the logic, functions, classes, and data.
Graphical User Interface	GUI	A visual user interface that allows the user to interact with the program through graphical icons and visual indicators.
User Interface	UI	The visual and interactive design of a software application that enables users to engage and interact with its features and content.
GUI Frame	-	A container that helps organize the GUI into different sections.

2. Product Scope

The product scope of the SMRT Vocab application is to help users expand their vocabulary of foreign languages through digital flashcards. The application shall allow users to study words in various languages, track their progress, and reinforce learning with spaced repetition. Key features include studying from flashcards (multiple-choice answer selection or text-based response), support for learning multiple foreign languages, text-to-speech pronunciation, and customizable study settings. The primary focus is on enhancing vocabulary acquisition for language learners of all levels.

Out of scope for this product are advanced language learning features such as grammar lessons, conversational practice, or full course material integration.

3. System Design Decisions and Constraints

3.1 System Design Decisions

System Design Decision ID	Date	Design Decision	Reason
DD-1	10/22/24	The application shall be a desktop application.	This application is an application that a user should focus on to retain more information. The user should be focused and distraction free. Therefore, increasing progression in the language of choice.
DD-2	10/22/24	The application shall use icons in addition to text to simplify the user interface.	The application users have varying ages including teenagers and elders. To make the application user-friendly, the icons will help to recognize actions in the buttons quickly.
DD-3	10/22/24	The application shall use a database to store user data.	The user data will be needed across use sessions. Having a database will ease to extract necessary data without the need to extract all data related to a user.
DD-4	10/22/24	The application shall use Python libraries for graphical interfaces.	The application includes many user interfaces. Python libraries natively support an easy-to-use graphical package. Using these libraries will speed up development.
DD-5	10/22/24	The application shall use simple user interfaces.	The application shall use graphical user interfaces designed with simplicity as the main driving force to facilitate ease of use.

3.2 System Design Constraints

System Design Constraint ID	Date	Constraint	Reason
DC-1	10/24/2025	The application shall be developed with Python.	The Python programming language is proven to be inefficient in multiple aspects. Due to the lack of focus on memory management principles and other optimizations the language of choice adds substantial weight to the system.
DC-2	10/24/2025	The application shall have a text based, comma separated value database.	The application is expected to manage moderately large word sets and individual users progress with understanding the word sets. CSV is effective and light in this regard but reduces functionality and augmentation of the database

DC-3	10/24/2025	The application is only ported for desktop PC and laptop use.	Due to the focused nature of the application the lack of a mobile application will limit the portability of the learning experience for users.
DC-4	10/24/2025	The application shall not use more than 4 GB of memory space.	The average desktop RAM is 8 GB. The most common operating system Windows 11 uses 4 GB of memory. Therefore, the application should work with a maximum of 4 GB of memory.

4. Software Architectural Design

The software architecture of the SMRT Vocab application is described in this section.

5. Software Design - Modules/Classes

Module/Class Name	Module Short Name (If there is one)	Short Description of the Module/Class
Main	Main	This class is responsible for creating the application window and managing the major GUI frames including MenuGUI, ChoiceGUI, TextGUI, SignupGUI, LoginGUI, SettingsGUI, and StatsGUI.
MenuGUI	MenuGUI	This class is responsible for handling the main menu, which provides buttons for the user to access ChoiceGUI, TextGUI, StatsGUI, and SettingsGUI. The menu also allows users to log out.
ChoiceGUI	ChoiceGUI	This class implements the multiple-choice flashcards.
TextGUI	TextGUI	This class implements the text-input flashcards.
ReviewGUI	ReviewGUI	This class implements the review mode of the application which uses a slightly different learning algorithm and prompts the user with a combination of multiple choice and text input flaschards.
LoginGUI	LoginGUI	This class allows users to login by providing a username and password. This class will also fetch the user's data if they were a previous user. This class allows users to create an account and sign up with an email and password.
StatsGUI	StatsGUI	This class is responsible for retrieving user data and displaying a statistical summary.
SettingsGUI	SettingsGUI	This class is responsible for the settings menu, which allows users to change how many correct flashcard attempts are required for a word to be marked as known. It also has settings for adjusting the size of the Walking Window and spaced repetition queue, whether the user wants to be prompted with foreign or English words, and which language the user wants to study.
Settings	Settings	This class contains the global variables that the SettingsGUI modifies so that the other modules can access them.
About	About	This class contains the frame for the about page of the applications which users may read.

Word	Word	This class contains and manipulates information about a user's word. This includes translations, the number of times the user has seen the word, the number of times the user has correctly translated the word, the number of times the user has incorrectly translated the word, and if the user knows the word.
WalkingWindow	WalkingWindow	This class contains and controls the Word objects that the user is currently learning. It reads in words from a CSV file and manipulates the Word objects as they are studied by the user. It implements spaced repetition using a FIFO queue.

5.1 Main

This class serves as the primary driver class or driver file for the application. Running this class results in execution of the application and all other associated classes and files. The GUI class is defined under this file

Attributes:

- App & ctk: these attributes include the definitions of the general feel and appearance of all GUI objects. **This attribute is included in most classes.**
- Study Window: The Walking Window that contains and manages the Words that the user can study from.

Methods:

- Show Menu GUI: Initializes the window for the menu of the application.
- Show Choice GUI: Initializes the window for the Choice GUI.
- Show Text GUI: Initializes the window for the Text GUI
- Show Review GUI: Initializes the window for the Review GUI.
- Show Settings GUI: Initializes the window for the Settings GUI.
- Show Statistics GUI: Initializes the window for the Statistics GUI.
- Show About GUI: Initializes the window for the About GUI.
- Main Loop: This method holds all GUI elements in place until action events occur in which the GUI changes.

5.2 MenuGUI

This class defines the main menu which is the first GUI window to appear when user executes the application. It provides buttons for all associated classes and provides the basis for user experience and navigation. This class includes button functions leading to Choice GUI, Text GUI, Settings GUI.

Attributes:

- Frame: Uses inbuilt ctk functions to create frame appearance and behavior. his attribute is included in most GUI classes.
- Button Font: Defines the font, size, weight, and appearance of text within buttons.
- Choice Button: The button for entering the flashcard GUI.
- Text Button: The button for entering the text-input GUI.
- Review Button: The button for entering the review-mode GUI.
- Stats Button: The button for entering the statistics GUI.
- Settings Button: The button for entering the settings GUI.
- Statistics Button: The button for entering the statistics GUI.
- About Button: The button for entering the about GUI.
- Exit Button: The button to exit the application.

Methods:

- Choice Function: Displays the Choice GUI.

- Text Function: Displays the Text GUI.
- Settings Function: Displays the Settings GUI.
- Statistics Function: Displays the Statistics GUI.
- Review Function: Displays the Review GUI.
- About Function: Displays the Review GUI.
- Exit Button: Destroys the GUI window to exit the application.

5.3 ChoiceGUI

This class defines the multiple-choice-based flashcard GUI. This class is a major feature of the primary learning mode of the application. It retrieves a set of random words from the Walking Window and picks one of them for the user to study and uses the others as options for the multiple-choice flashcards.

Attributes:

- Flashword, var1, var2, var3: These variables store words at random for presentation to the user in the form of buttons.
- Choices: A list of words that stors the choices possible for the user to make.
- Buttons: A list of buttons as defined in which the variable flashcard options will be placed.
- Flashcard: ctk label to hold the word being trained.
- Known Word: This calls a function to set the known value of a word to 1 and remove it from study_window.
- Answer Position: Selects a random integer to be the correct answer of the multiple choice.
- Flashword: This gets a random word from the list of words the user is currently learning.
- Flash Font: Defines the appearance of text within the flashword provided.
- Button Font: Defines the appearance of text within the buttons on this page.

Methods:

- Back Function: This returns the user to Menu GUI class.
- Known Word Function: calls study window's set word known function.
- Display Feedback: Sends the chosen answer to the Walking Window to check if the definition is correct and displays feedback accordingly.

5.4 TextGUI

This class defines the inputs and outputs of the text-input based flashcards that are used for the learning mode. This class is an alternative version of the Choice GUI except with only a word and a textbox for user input and validation.

Attributes:

- Back Button Font: Defines the appearance of the back button.
- Known Word: This calls a function to set the known value of a word to 1 and remove it from study_window.
- Flashcard: A label used to present a word for the user to translate.
- Input Frame: Frame to hold the text entry box.
- Text Entry: This frame is a ctk object which is where the user will type their answer to the prompt.

Methods:

- Back Function: Returns user to Menu GUI class.
- Known Word Function: calls study window's set word known function.
- Display Feedback: Sends the entered answer to the Walking Window to check if the definition is correct and displays feedback accordingly.

5.5 LoginGUI

This class allows users to register for the application and log in to the application by providing a username and password.

Attributes:

- Frame: Uses inbuilt ctk functions to create frame appearance and behavior. his attribute is included in most GUI classes.
- Username Entry: An input field for the user's username.
- Password Entry: An input field for the user's password.
- Login Button: A button to submit the login form.
- Signup Button: A button to submit the signup form.
- Exit Button: A button to exit the application.
- Welcome Label: A label to display a friendly welcome message.
- Feedback Label: A label to display messages (e.g., "Login failed" or "Please enter both username and password").

Methods:

- Init: Initializes the GUI components and sets up the login form.
- Switch Welcome Message: Displays a welcome message that rotates between several languages.
- Login Function: Processes the login attempt, such as sending the credentials to a server or checking against a local database. Will create the necessary CSV files if needed.
- Signup Function: Creates a user account for the provided credentials and logs the user in to the application. Will create the necessary CSV files if needed.
- Show Feedback: Displays feedback messages to the user, such as indicating if the login was successful or if there was an error.

5.6 StatsGUI

This class is the portion of the GUI that displays the user’s learning statistics. This class displays the user’s 5 worst-known words, 5 best-known words, number of known words, and an estimated language comprehension percentage.

Attributes:

- Interactive Graph: Visual representation of the users learning and progress which the user can also set goals with the slider function.
- Most Seen Words: The words that the user has seen the greatest number of times the greatest number of times.
- Worst Words: The words that the user has translated incorrectly the greatest number of times.
- Known Words: the count of words that are marked as known for the user.

Methods:

- Get Best Words: retrieves the user’s word data then finds and returns the 5 words that were translated correctly the most.
- Get Worst Words: retrieves the user’s word data then finds and returns the 5 words that were translated incorrectly the most.
- Get Known Words: retrieves the user’s word data then finds and returns the number of words marked as known.

5.7 SettingsGUI

This class is responsible for the settings menu, which allows users to change how many correct flashcard attempts are required for a word to be marked as known. It also has settings for adjusting the size of the Walking Window and spaced repetition queue, whether the user wants to be prompted with foreign or English words, and which language the user wants to study.

Attributes:

- Known Threshold: The number of correct translations needed for a word to be marked as known
- Known Delta: The difference between correct and incorrect attempts to learn a word to be marked as

known.
- SRS Queue Size: The size of the buffer for testable correct words (How often you will see correct words)
- Walking Window Size: The size of the Walking Window as the number of words in the Walking Window.
- Username: The user's username, used to find the user's word data.
- Foreign to English: A Boolean value designating if the user wants to be prompted with foreign or English words to translate.
- Language: The foreign language that the user wants to study.

Methods:
- Add Slider: Creates a slider for one of the settings.
- Update Slider Label: Updates the label on the sliders to match their current value.
- Update Toggle Labl: Updates the label on the toggle to match its current value.
- Save Settings: Saves the current values of the settings to the Settings module.

5.8 Settings

This class contains the global variables that the SettingsGUI modifies so that the other modules can access them.

Attributes:
- Known Threshold: The number of correct translations needed for a word to be marked as known.
- Known Delta: The difference between correct and incorrect attempts to learn a word to be marked as known.
- SRS Queue Size: Size of the buffer for testable correct words (How often you will see correct words)
- Walking Window Size: The size of the Walking Window as the number of words in the Walking Window.
- Username: The user's username, used to find the user's word data.
- Foreign to English: A Boolean value designating if the user wants to be prompted with foreign or English words to translate.
- Language: The foreign language that the user wants to study.

5.9 Word

This class contains the data and behavior associated with a single word for one user.

Attributes:
- English: The English translation of the word.
- Foreign: The foreign translation of the word.
- Seen Count: The count of how many times the user has been prompted to translate the word.
- Correct Count: The count of how many times the user correctly translated the word.
- Incorrect Count: The count of how many times the user incorrectly translated the word.
- Known: is a Boolean value that marks if the user knows the word.

Methods:
- Check Definition English: Checks the input string variable for equivalency with the correct English version of the Word object that the method is called from.
- Check Definition Foreign: Checked the input string variable for equivalency with the correct foreign version of the Word object that the method is called from.
- Correct: Increments the count seen and count correct attributes.
- Incorrect: Increments the count seen and count incorrect attributes.
- Set Known Word: Sets the known attribute to true.
- Check If Known: Checks if the word is known based off the values in the Settings module.

5.10 WalkingWindow

This class contains and controls the Word objects that the user is currently learning. It reads in words from a CSV

file and manipulates the Word objects as they are studied by the user. It implements spaced repetition using a FIFO queue.

Attributes:

- Size: The number of words in the Walking Window.
- SRS Queue: The FIFO queue that is used to implement spaced repetition.
- Current Words: The words currently in the Walking Window that can be chosen to study.
- Words Dictionary: A dictionary used to store the user's study progress on each word.

Methods:

- Init: This method is the constructor of the class. It is automatically called when a new instance of the class is created. It initializes the class's attributes.
- CSV To Words Dictionary: This method loads a list of words from a CSV file into the Words Dictionary. The CSV file acts as the data source, containing the words the user wants to learn.
- Get Random Words: This method returns a random selection of words from the loaded list. It is used for practicing or testing knowledge of words in a randomized order.
- Init Current Words: Initializes the Walking Window with a number of words that are not yet known.
- Check Word Definition: Checks if a Word was translated correctly and processes the result. Moves Words into the SRS Queue if they are correctly translated or removes them from the Walking Window if they are now known.
- Remove Known Word: This method removes a word from the walking window that becomes known via testing.
- Add New Word: This method pulls the next word to be studied from the CSV.

5.11 TestToSpeech

This class pulls audio from the gtts library and presents it to the Test, Choice, and Review GUIs in order to allow a text to speech button functionality. TestToSpeech pulls .mp3 files and then stores them locally in the system under audio file directories to allow rapid recall of already downloaded audio.

Attributes:

- Base Audio Directory: This attribute defines where audio files are stored within the program.
- Languages: This determines which languages are available, identifies and uses them as variables for naming conventions and logic.
- Audio file path: a more specialized path to the specific language's audio files.

Methods:

- Get Audio File: This method takes a word from a specified language and fetches the .mp3 file for that word.
- Generate Pronunciation: This method provides the majority of the logic from the word class and the Text to Speech class and implements them with the gtts library function to provide audio files to then be saved and sent to Play Audio Async.
- Play Audio Async: This method provides the actual audio output of the mp3 file whenever a user clicks the text to speech buttons in the various GUIs.

6. User Interface Design

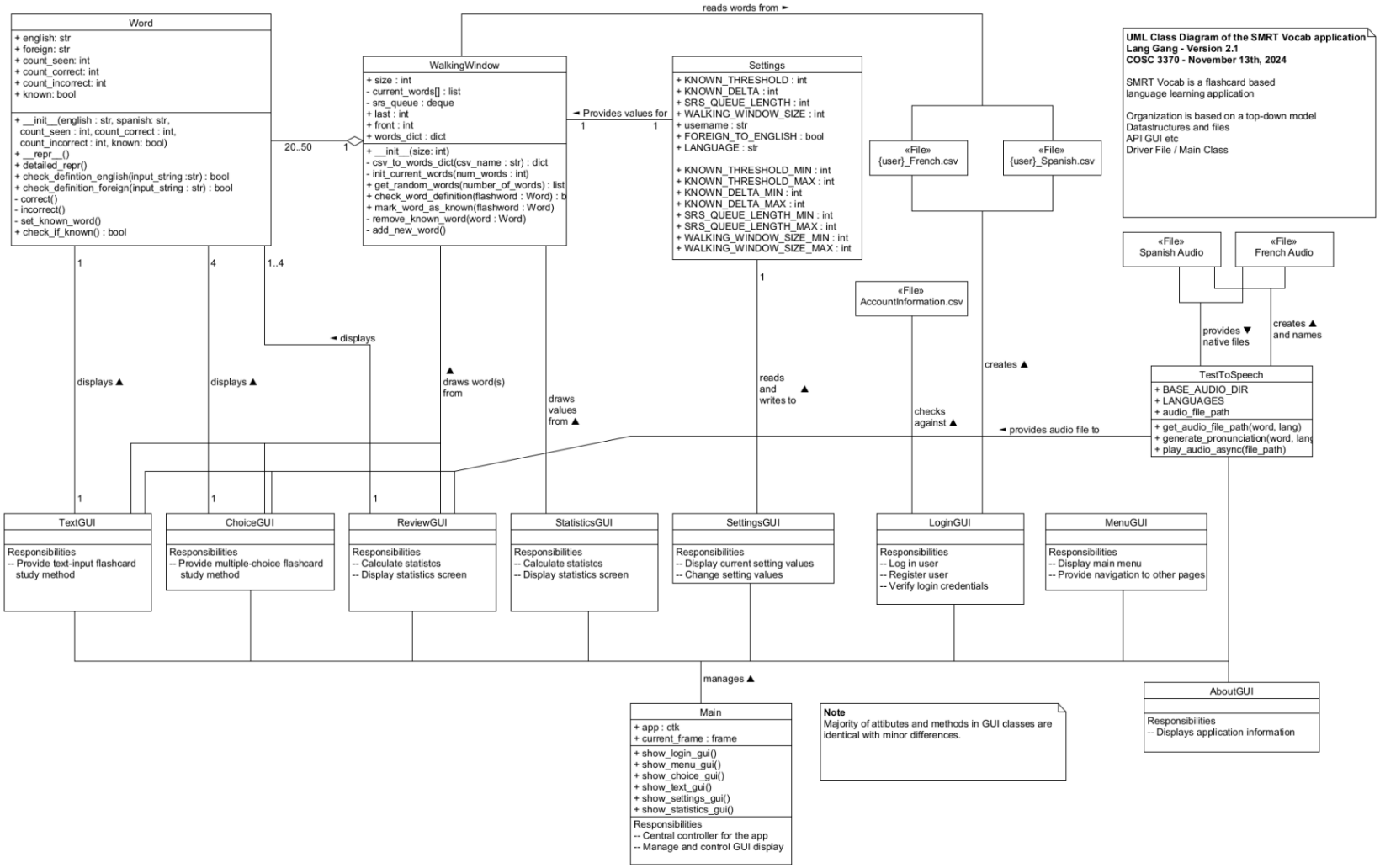
The user interface of SMRT Vocab shall prioritize simplicity, intuitiveness, and efficiency to encourage users' engagement and learning. Following are the descriptions of the UI designs of the application:

- User Login: The SMRT Vocab login screen is designed for quick, easy access with a clean, inviting layout featuring the app's logo and a welcoming tagline. Users can sign in via email, with a "Sign Up" option for

new users. The minimalistic design enhances accessibility and encourages users to engage immediately with the app.

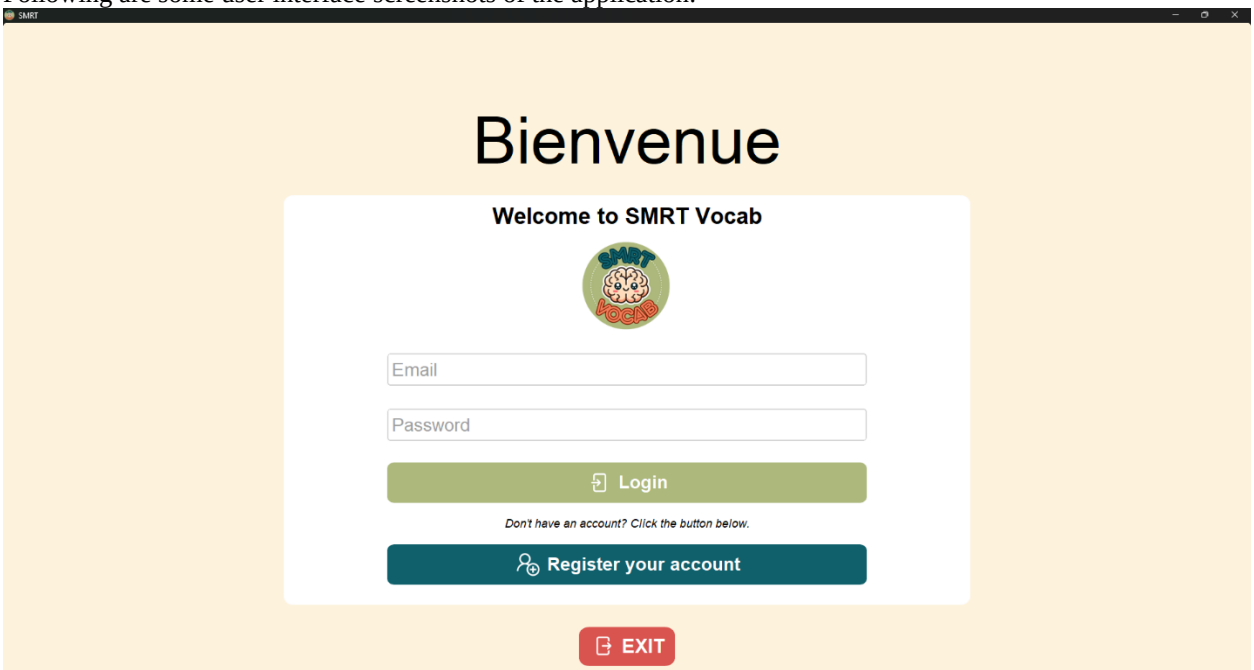
- Main Menu Screen: The main menu screen of SMRT Vocab offers navigation choices between Multiple-Choice Flashcards, Text-Input Flashcards, Statistics, and Settings. Each option is displayed as a bold, easily tappable button with clear icons, allowing users to quickly navigate to their preferred activity. The clean, intuitive design and simple icons make it easy for users to jump right into learning or track their progress.
- Multiple-Choice Flashcards: The flashcard screen in SMRT Vocab features a clean, focused design, displaying one word at a time. Users are prompted to select the correct word from a selection of answer choices.
- Text-Input Flashcards: The text-input flashcard screen provides users with a single word to manually translate to reinforce vocabulary retention. Each exercise is displayed in a simple, uncluttered format with a single word and text-input box, and immediate feedback is given after each response to aid learning. This layout encourages users to focus on comprehension.

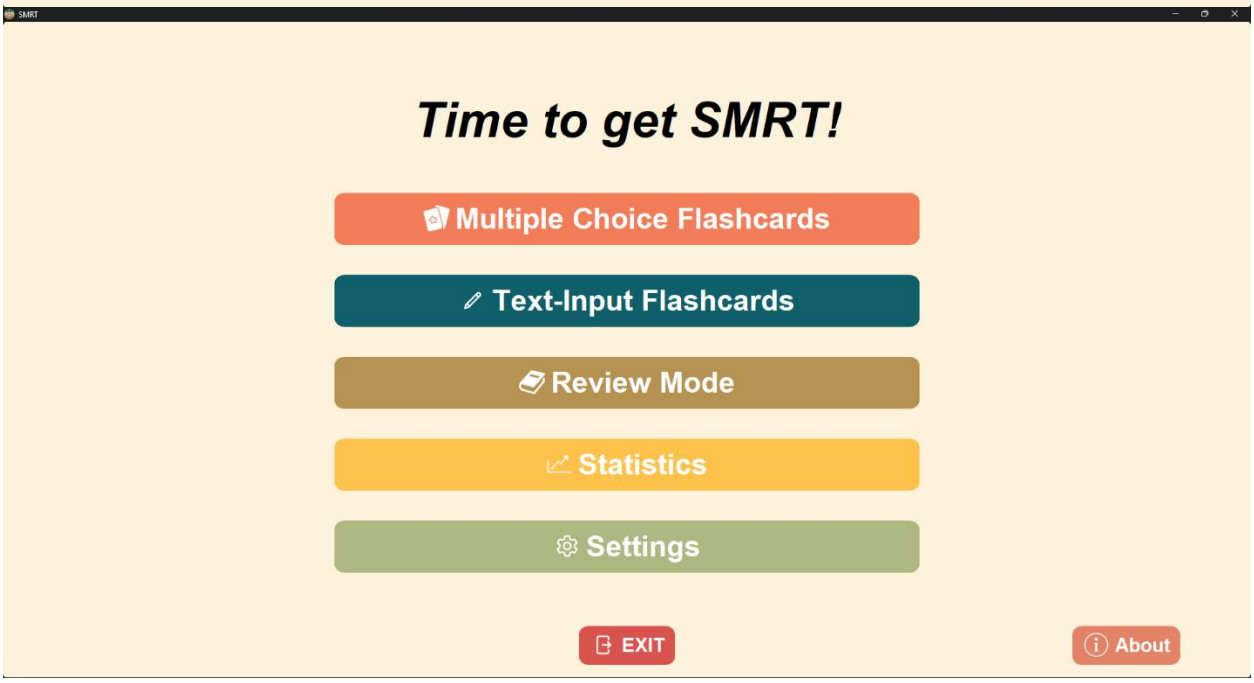
7. Appendix A – Class Diagram of the Software System

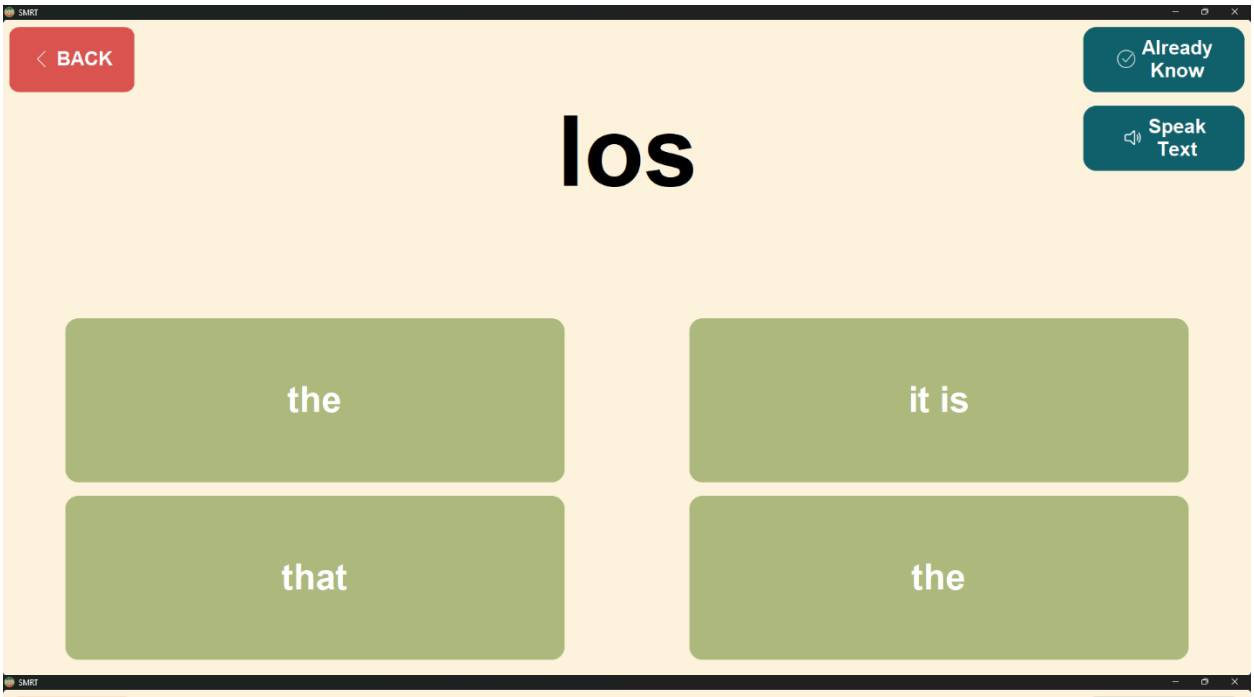


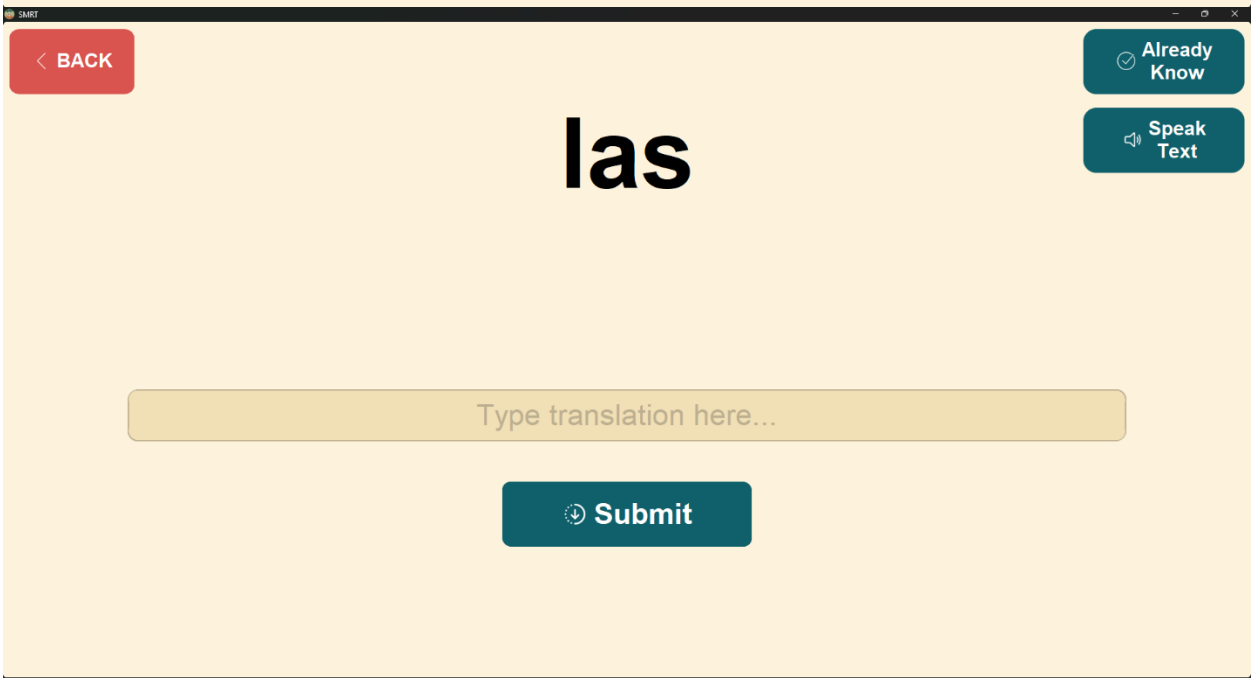
8. Appendix B – User Interfaces

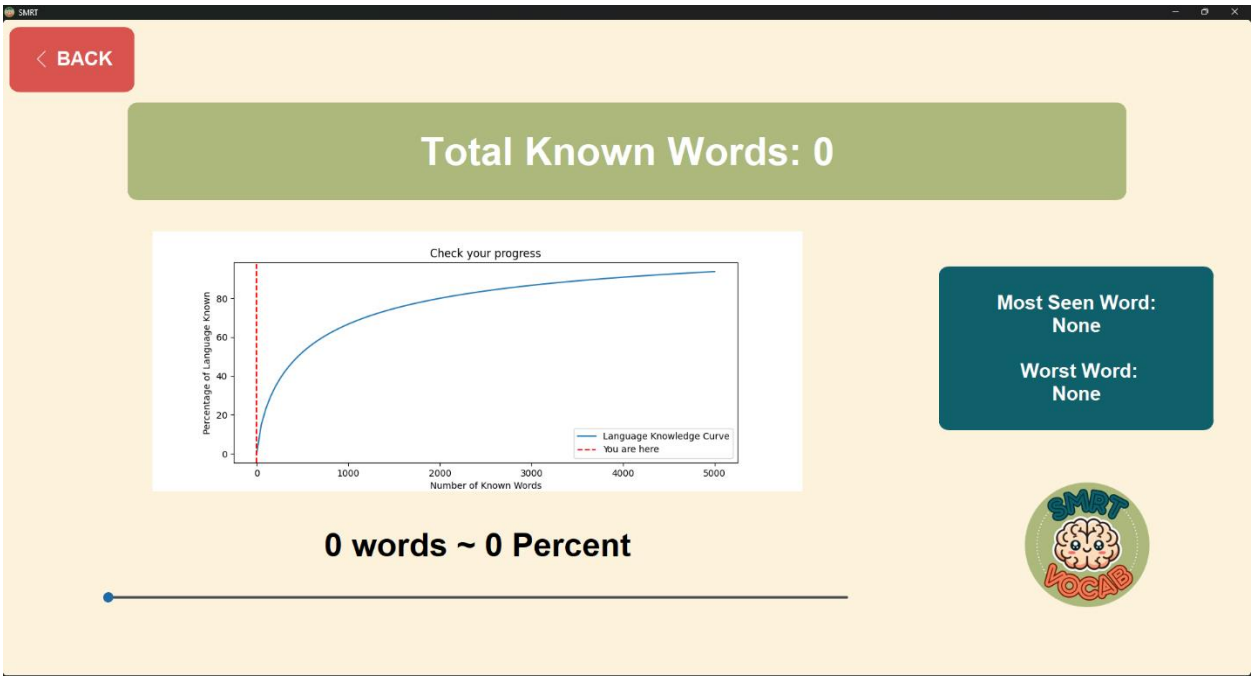
Following are some user interface screenshots of the application:

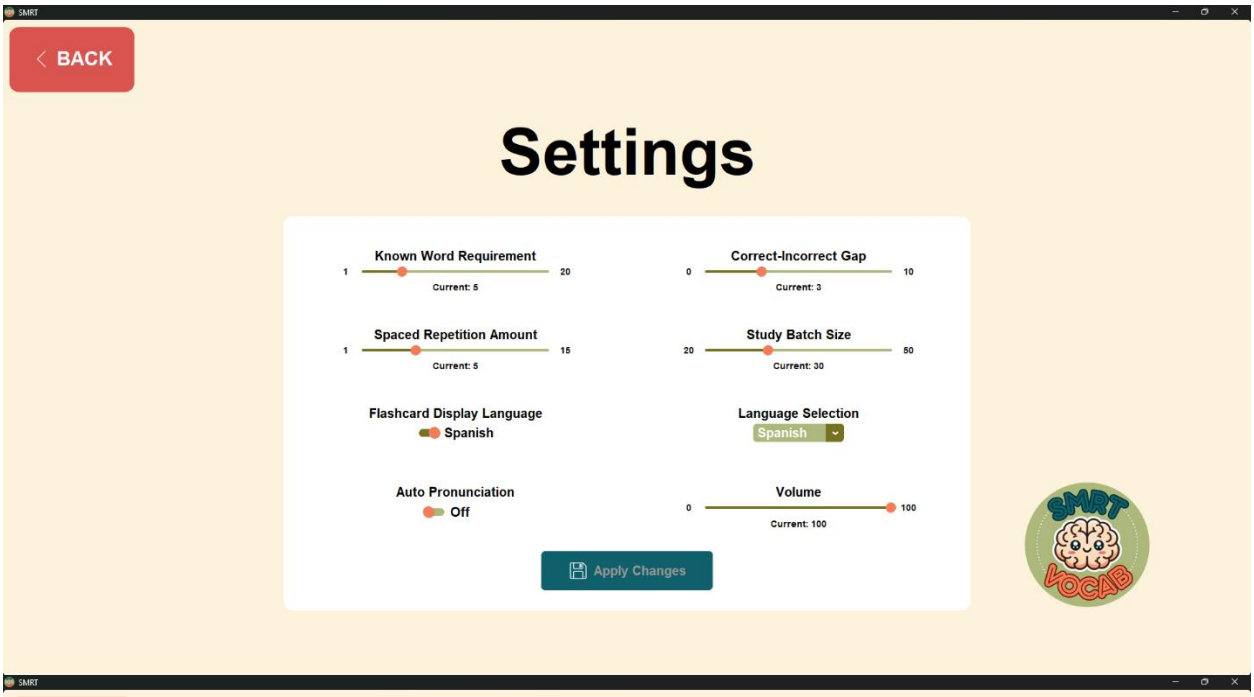


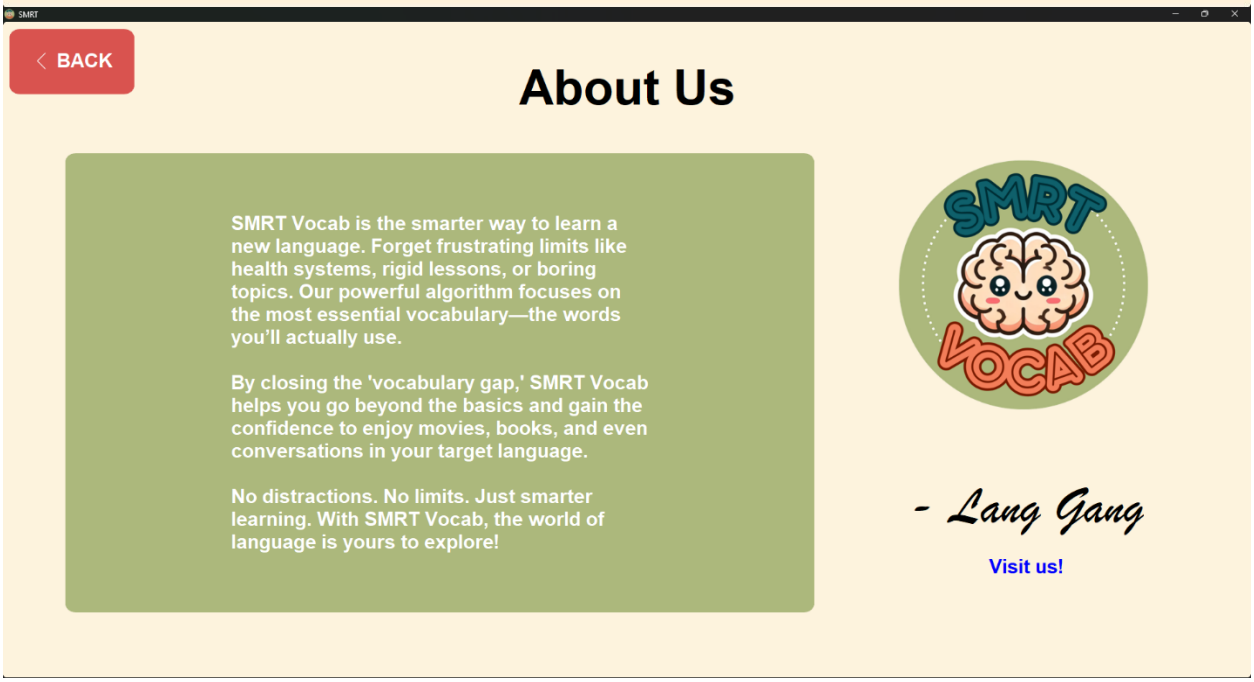



BACK
Already Know
Speak Text
los
the
it is
that
the

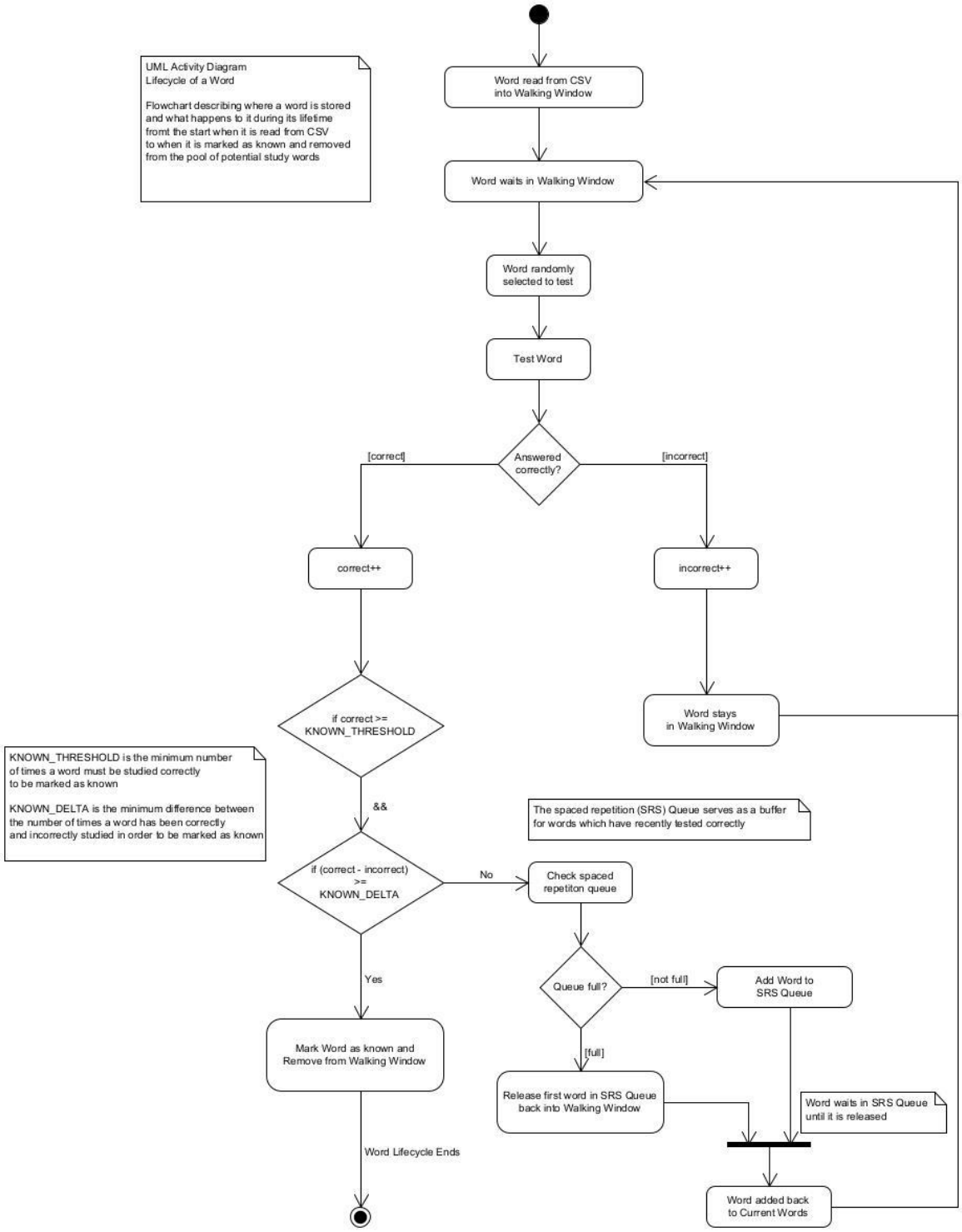

BACK
Already Know
Speak Text
las
Type translation here...
Submit


SMRT
BACK
Total Known Words: 0
Check your progress
Percentage of Language Known
Number of Known Words
Language Knowledge Curve
You are here
Most Seen Word:
None
Worst Word:
None
0 words ~ 0 Percent


BACK
Settings
Known Word Requirement
1
20
Current: 5
Correct-Incorrect Gap
0
10
Current: 3
Spaced Repetition Amount
1
15
Current: 5
Study Batch Size
20
50
Current: 30
Flashcard Display Language
Spanish
Language Selection
Spanish
Auto Pronunciation
Off
Volume
0
100
Current: 100
Apply Changes
SMRT
VOCAB


BACK
About Us
SMRT Vocab is the smarter way to learn a new language. Forget frustrating limits like health systems, rigid lessons, or boring topics. Our powerful algorithm focuses on the most essential vocabulary—the words you'll actually use.
By closing the 'vocabulary gap,' SMRT Vocab helps you go beyond the basics and gain the confidence to enjoy movies, books, and even conversations in your target language.
No distractions. No limits. Just smarter learning. With SMRT Vocab, the world of language is yours to explore!
SMRT
VOCAB
- Lang Gang
Visit us!

9. Appendix C – UML Activity Diagram Lifecycle of a Word



Lang Gang

SMRT Vocab
Software Test Description (STD)

Version 2.0
11/21/24

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
There are an adequate number of test cases to cover all the necessary testing for the application.	YES
The test traceability matrix includes all the test cases and all application requirements including functional, hardware, and quality.	YES
All test cases in the test case list are detailed in the appendixes.	YES
The planned test duration is adequately estimated for the test case.	YES
All fields in the test cases are completed and no field is left empty.	YES
The test steps in the test case are detailed enough to adequately test the feature/s and the requirements.	YES
The test cases include at least 10 steps that are detailed enough to replicate the test with inputs and outputs.	YES

Final Control	**Yes/No**
The document is checked based on the "Team Project Artifacts Common Errors" Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Ryleigh Sibley	Project Technical Manager	Ryleigh Sibley	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Voss Purkey	Project Manager	Voss Purkey	11/21/24
Garrett Strader	Project Design Manager	Garrett Strader	11/21/24

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Garrett Strader	Project Design Manager	Garrett Strader	11/21/24
Jacob Poland	Project Configuration Manager	Jacob Poland	11/21/24
Zachary Kao	Project Test Manager	Zachary Kao	11/21/24
Malak Mahdy	Project Requirements Manager	Malak Mahdy	11/21/24
Ryleigh Sibley	Project Manager	Ryleigh Sibley	11/21/24
Voss Purkey	Project Manager	Voss Purkey	11/21/24
Marcelo Barahona	Software Engineer	Marcelo Barahona	11/21/24

Revision History

Date	Version	Description	Author
11/01/24	1.0	Initial iteration.	Ryleigh Sibley, Zachary Kao, Jacob Poland, Voss Purkey, Marcelo Barahona and Garrett Strader
11/10/24	1.1	This revision added authors in the revision history for the 1.0 release of this document and fixed a couple small formatting issues, and changed the checklist response regarding the number of steps per test case to "NO." Test case 1 was also updated.	Jacob Poland
11/12/24	1.2	Applied justified formatting to sections that were not properly formatted. Product scope updated to match other documents Updated references to refer to v2.0 of internal documents. Removed TC-2 and integrated it into TC-1. Removed TC-10 and integrated it into TC-2. Removed TC-8 and TC-9 and integrated them into TC-3 and TC-4. Removed TC-7 and integrated it into TC-6. Updated the traceability matrix to reflect the consolidated test cases.	Zachary Kao
11/12/24	1.4	Updated the priority levels of the test cases to be: Low- TC-5, Medium – TC-1, TC-6, High- TC-2, TC-3, TC-4.	Ryleigh Sibley
11/20/24	1.5	Updated TC-1.	Jacob Poland
11/21/24	1.6	Updated TC-4 and TC-5	Voss Purkey
11/21/24	1.7	Updated test schedule	Zachary Kao
11/21/24	2.0	Final check for submission	Malak Mahdy Marcelo Barahona

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Software Test Description

1. Introduction

1.1 Purpose of the Document

The Software Test Description (STD) outlines the testing activities and test cases for SMRT Vocab. Its purpose is to guide testers and team members in strategizing a comprehensive testing approach, ensuring that the software is thoroughly verified. This document details specific roles and responsibilities of stakeholders, outlines requirements for the testing environment, and provides guidelines for conducting tests. Additionally, the STD serves as a communication tool, keeping stakeholders informed about testing activities throughout the project. Approval of this document by stakeholders, especially the customer or user representative identified in the project plan, is required to proceed.

1.2 Scope of the Document

The Software Test Description (STD) outlines the scope of SMRT Vocab's testing process. It includes a high-level overview of the product's scope, defines stakeholder responsibilities related to testing activities, and presents the planned test schedule. Additionally, this document contains a comprehensive list of test cases, a traceability matrix linking test cases to requirements, and detailed descriptions of each test case to ensure thorough verification of the software.

1.3 References

Reference Title/Name	Date	Version
Project Description Document (PDD)	08/15/2024	2.0
Project Vision Document (PVD)	11/18/2024	3.0
Project Plan (PP)	11/18/2024	2.0
Software Requirements Specification (SRS)	11/18/2024	2.0
Software Design Description (SDD)	11/18/2024	2.0
MIL-STD-498	12/5/1994	N/A
IEEE/ISO/IEC 29148-2018 - ISO/IEC/IEEE International Standard - Systems and software engineering -- Life cycle processes -- Requirements engineering	10/23/2018	2018
Web Content Accessibility Guidelines (WCAG)	11/11/2008	2.0

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Project Description Document	PDD	The project description document is the project document detailing the project to be completed.
Project Vision Document	PVD	The project vision document is the project document describing the project vision to be accomplished.

Project Plan	PP	The project plan is the project document detailing the plan to be followed throughout the project.
Software Requirements Specification	SRS	The software requirements specification is the project document specifying the project requirements and the use cases to be satisfied for the successful completion of the project.
Software Design Description	SDD	The software design description is the project document specifying the project requirements and the use cases to be satisfied for the successful completion of the project.
Software Test Description	STD	The software test description is the project document specifying the test-related planning, activities, and test cases to successfully verify the software under development.

2. Product Scope

The product scope of the SMRT Vocab application is to help users expand their vocabulary of foreign languages through digital flashcards. The application shall allow users to study words in various languages, track their progress, and reinforce learning with spaced repetition. Key features include studying from flashcards (multiple-choice answer selection or text-based response), support for learning multiple foreign languages, text-to-speech pronunciation, and customizable study settings. The primary focus is on enhancing vocabulary acquisition for language learners of all levels.

Out of scope for this product are advanced language learning features such as grammar lessons, conversational practice, or full course material integration.

3. Test Planning

3.1 Responsibilities

Role	Name	Responsibilities
Test Engineer	Marcelo Barahona Garrett Strader	The test engineer is responsible for preparing the test cases under the direction of the test manager.
Developer	Jacob Poland	The developer supports the test engineers during the tests. They fix the defects/bugs if identified during tests. The developer remains actively involved throughout the testing process.
Test Manager	Zachary Kao	The test manager oversees all aspects of project testing, including planning and coordinating testing activities. They are responsible for taking corrective actions when necessary and regularly updating the project manager and technical manager on testing progress.
Customer/User Representative	Voss Purkey	The customer/user representative approves the software test description, reviews test cases, and supports the testing process if possible.
Requirements Manager	Malak Mahdy	The requirements manager approves the Software Test Description, reviews test cases, and supports the testing

		process if possible.
Technical Manager	Ryleigh Sibley	The technical manager is responsible for ensuring that all teams (project team members) work in tandem to achieve a successful testing process.

3.2 Test Schedule

Test Case	Scheduled Date	Actual Date / Duration
TC-1	11/14/2024	11/20/2024 : 20 minutes
TC-2	11/14/2024	11/20/2024 : 10 minutes
TC-3	11/14/2024	11/20/2024 : 15 minutes
TC-4	11/14/2024	11/20/2024 : 20 minutes
TC-5	11/14/2024	11/20/2024 : 20 minutes
TC-6	11/15/2024	11/20/2024 : 20 minutes

4. Test Preparation

4.1 Test Environment

The testing for SMRT Vocab will be conducted on a personal computer running the Windows 11 operating system. The application has been properly downloaded and installed on this machine, per the installation manual. Testing will primarily take place in a controlled laboratory environment or designated testing workspace, equipped with the essential hardware. No special equipment or software is required to complete these tests.

5. Test Cases

5.1 Test Cases

Test Case ID	Test Case Name	Description
TC-1	Registration and Login Test	This test case verifies various successful and unsuccessful user registration and login scenarios.
TC-2	Menu Navigation Test	This test case verifies that menu navigation to and from each page is successful and timely when run on the suggested hardware.
TC-3	Multiple-Choice Flashcard Test	This test case verifies various correct and incorrect inputs to the multiple-choice user interface as well as verifies the text-to-speech, mark as known, and data saving features.
TC-4	Text-Input Flashcard Test	This test case verifies various correct and incorrect inputs to the text-input user interface as well as verifies the text-to-speech, mark as known, and data saving features.
TC-5	Review Mode Test	This test case verifies that the Review Mode feature properly presents a random known word to the review in either a multiple-choice or text-input flashcard, at random.
TC-6	Change Settings and View	This test case verifies various changes to the options in the

	Statistics Test	settings menu and that the statistics page is correctly showing the user's statistics.

6. Appendix A – Test Case to Requirements Traceability

Test Case ID	Requirement ID
TC-1	Req-Func-Sw-1
TC-2	Req-Func-Sw-5 Req-Func-Hw-1 Req-Func-Hw-2 Req-Func-Hw-3 Req-Func-Hw-4 Req-Qual-Perf-1 Req-Qual-Use-2
TC-3	Req-Func-Sw-2 Req-Func-Sw-4 Req-Func-Sw-5 Req-Func-Sw-6 Req-Func-Sw-12 Req-Qual-Perf-2 Req-Qual-Use-1
TC-4	Req-Func-Sw-3 Req-Func-Sw-4 Req-Func-Sw-5 Req-Func-Sw-6 Req-Func-Sw-12 Req-Qual-Perf-2 Req-Qual-Use-1
TC-5	Req-Func-Sw-2 Req-Func-Sw-3 Req-Func-Sw-4 Req-Func-Sw-5 Req-Func-Sw-10 Req-Qual-Perf-2 Req-Qual-Use-1
TC-6	Req-Func-Sw-5 Req-Func-Sw-8 Req-Func-Sw-9 Req-Func-Sw-10 Req-Func-Sw-11 Req-Qual-Scal-1

7. Appendix B – Test Case 1 – Registration and Login Test

Test Case ID	TC-1
Test Case Name	Registration and Login Test
Brief Description	This test case verifies various successful and unsuccessful user registration and login scenarios.
Priority	Medium

Planned Test Duration	30 minutes
Current Version	2.1
Date of First Version	10/31/2024
Date of Current Version	11/12/2024
Created By	Garrett Strader
Last Update By	Jacob Poland
Approved By	Ryleigh Sibley
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Jacob Poland
List of Test Support Software	N/A
List of Test Support Hardware	N/A

Test Case 1 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running.	Pass
2	Click “Register your account.”	The application displays a feedback box that says, “Please Complete All Fields” and an OK button.	Pass
3	Click OK.	The feedback box and OK button disappear.	Pass
4	Enter an email that is already in the accounts table. Click “Register your account.”	The application displays a feedback box that says, “Please Complete All Fields” and an OK button.	Pass
5	Click OK and enter a password and click, “Register your account.”	The application displays a feedback box that says, “That email is already in use! Please register with another email”	Pass
6	Click OK and enter an email that contains any of the following unallowed characters: <>:"/\\\|?*	The application displays an OK button and a feedback box that says, “Emails should not contain the following characters: <>:"/\\\|?*.”	Pass
7	Click OK and remove the	The application displays	Pass

	forbidden character(s) from the email. The email should not already be registered to an account. Click, "Register your account."	"Registration Success" and then the main menu.	
8	Click Exit.	The application closes.	Pass
9	Run the application again.	The application starts running.	Pass
10	Click Login.	The application displays a feedback box displaying "Please Complete All Fields".	Pass
11	Enter an email that was previously registered. Enter a password that does not match the registered email. Click Login.	The application displays a feedback box displaying "Incorrect Email or Password".	Pass
12	Click OK and enter the correct password. Click Login.	The application displays the main menu.	Pass

8. Appendix D – Test Case 2 – Menu Navigation Test

Test Case ID	TC-2
Test Case Name	Menu Navigation Test
Brief Description	This test case verifies that menu navigation to and from each page is successful and timely when run on the suggested hardware.
Priority	High
Planned Test Duration	10 minutes
Current Version	2.0
Date of First Version	10/31/2024
Date of Current Version	11/12/2024
Created By	Jacob Poland
Last Update By	Zachary Kao
Approved By	Ryleigh Sibley
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Malak Mahdy
List of Test Support Software	N/A
List of Test Support Hardware	N/A

Test Case 2 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running within 3 seconds.	Pass
2	Log in to the application.	The application displays the main menu within 3 seconds.	Pass
3	Click the Multiple-Choice Flashcards button.	The application displays a flash card with multiple translation choices and a Back button within 3 seconds.	Pass
4	Click the Back button.	The application displays the main menu within 3 seconds.	Pass
5	Click the Text-Input Flashcards button.	The application displays a flashcard with a text-input box, a Submit button, and a Back button within 3 seconds.	Pass
6	Click the Back button.	The application displays the main menu within 3 seconds.	Pass
7	Click the Review button.	The application displays the Review mode menu within 3 seconds.	Pass
8	Click the Back Button.	The application displays the main menu within 3 seconds.	Pass
9	Click the Statistics button.	The application displays the user's statistics and a Back button within 3 seconds.	Pass
10	Click the Back button.	The application displays the main menu within 3 seconds.	Pass
11	Click the Settings button.	The application displays the user's settings and a Back button within 3 seconds.	Pass
12	Click the Back button.	The application displays the main menu within 3 seconds.	Pass
13	Click the About button.	The application displays an About description and a link to the SMRT Vocab website within 3 seconds.	Pass
14	Click the Back button.	The application displays the main menu within 3	Pass

		seconds.	
15	Click the Exit button.	The application window closes.	Pass

9. Appendix E – Test Case 3 – Multiple-Choice Flashcard Test

Test Case ID	TC-4
Test Case Name	Multiple-Choice Flashcard Test
Brief Description	This test case verifies various correct and incorrect inputs to the multiple-choice user interface as well as verifies the text-to-speech, mark as known, and data saving features.
Priority	High
Planned Test Duration	1 hour
Current Version	2.0
Date of First Version	10/31/2024
Date of Current Version	11/12/2024
Created By	Voss Purkey
Last Update By	Zachary Kao
Approved By	Ryleigh Sibley
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Zachary Kao
List of Test Support Software	N/A
List of Test Support Hardware	N/A

Test Case 3 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running.	Pass
2	Login to the application.	The application displays the main menu.	Pass
3	Click the Multiple-Choice Flashcards button.	The application presents GUI and flashcard options as expected.	Pass
4	Click an incorrect answer to the flashcard prompt.	The application displays a feedback box notifying the user that they chose the	Pass

		incorrect answer and shows the correct answer and an OK button. This feedback should be displayed in under 0.5 seconds.	
5	Click the flashcards behind the feedback window.	The application does not process the button press. No additional word progress is logged to smrt.log.	Pass
6	Click OK.	The feedback box disappears and a new flashcard and set of answers is displayed.	Pass
7	Click the Text-to-Speech button.	The application plays an audio pronunciation of the displayed word.	Pass
8	Click the Already Known button.	The application marks the currently displayed word as known and logs this action to smrt.log. The application displays a new flashcard and set of answers.	Pass
9	Click the correct answer to the flashcard prompt.	The application displays a feedback box with a message that the user chose the correct answer. This feedback should appear within 0.5 seconds and stay on the screen for 1 second.	Pass
10	Exit the application.	The application closes.	Pass
11	Check smrt.log.	The log should contain information confirming that the previous steps were successful.	Pass
12	Check the CSV file corresponding to the chosen user and language.	The CSV file should reflect the correct, incorrect, and marked as known words from the previous steps.	Pass

10. Appendix F – Test Case 4 – Text-Input Flashcard Test

Test Case ID	TC-4
Test Case Name	Text-Input Flashcard Test
Brief Description	This test case verifies various correct and incorrect inputs to the text-input user interface as well as verifies the text-to-speech, mark as known, and data saving features.
Priority	High
Planned Test Duration	1 hour

Current Version	2.0
Date of First Version	10/31/2024
Date of Current Version	11/12/2024
Created By	Voss Purkey
Last Update By	Zachary Kao
Approved By	Zachary Kao
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Marcelo Barahona
List of Test Support Software	N/A
List of Test Support Hardware	N/A

Test Case 4 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running.	Pass
2	Login to the application.	The application displays the main menu.	Pass
3	Click the Text-Input Flashcard button.	The application displays the text-input flashcard GUI as expected.	Pass
4	Enter the wrong answer into the text box and press Submit.	The application displays a feedback box notifying the user that they chose the incorrect answer and shows the correct answer and an OK button. This feedback should be displayed in under 0.5 seconds.	Pass
5	Click the submit button while the feedback box is shown.	The application does not process the button press. No additional word progress is logged to smrt.log.	Pass
6	Click OK.	The feedback box disappears and a new flashcard and set of answers is displayed.	Pass
7	Click the Text-to-Speech button.	The application plays an audio pronunciation of the displayed word.	Pass
8	Click the Already Known	The application marks the	Pass

	button.	currently displayed word as known and logs this action to smrt.log. The application displays a new flashcard and set of answers.	
9	Type the correct answer to the flashcard prompt and press Submit.	The application displays a feedback box with a message that the user chose the correct answer. This feedback should appear within 0.5 seconds and stay on the screen for 1 second.	Pass
10	Exit the application.	The application closes.	Pass
11	Check smrt.log.	The log should contain information confirming that the previous steps were successful.	Pass
12	Check the CSV file corresponding to the chosen user and language.	The CSV file should reflect the correct, incorrect, and marked as known words from the previous steps.	Pass

11. Appendix F – Test Case 5 – Review Mode Test

Test Case ID	TC-5
Test Case Name	Review Mode Test
Brief Description	This test case verifies that the Review Mode feature properly presents a random known word to the review in either a multiple-choice or text-input flashcard, at random.
Priority	Low
Planned Test Duration	1 hour
Current Version	1.0
Date of First Version	11/12/2024
Date of Current Version	11/12/2024
Created By	Zachary Kao
Last Update By	Zachary Kao
Approved By	Marcelo Barahona
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Jacob Poland
List of Test Support Software	N/A
List of Test Support	N/A

Hardware	

Test Case 5 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running.	Pass
2	Log in to the application with an account that has no known words.	The application displays the main menu.	Pass
3	Click on the Review Mode button.	The application displays a message notifying the user that they have no words to review.	Pass
4	Exit the application.	The application closes.	Pass
5	Run the application again.	The application starts running.	Pass
6	Log in to the application with the test account using the email “test@test.com” and password “test”.	The application displays the main menu.	Pass
7	Click on the Review Mode button.	The application displays the review mode screen and either the multiple-choice or text-input flashcard for a known word is displayed.	Pass
8	Answer review questions until at least one multiple-choice and one text-input is answered correctly.	The application displays a feedback box with a message that the user chose the correct answer. This feedback should appear within 0.5 seconds and stay on the screen for 1 second.	Pass
9	Answer review questions until at least one multiple-choice and one text-input is answered incorrectly.	The application displays a feedback box notifying the user that they chose the incorrect answer and shows the correct answer and an OK button. This feedback should be displayed in under 0.5 seconds.	Pass
10	Click the Text-to-Speech button.	The application plays an audio pronunciation of the displayed word.	Pass
11	Exit the application.	The application closes.	Pass
12	Check smrt.log.	The log should contain information confirming that the previous steps were successful.	Pass

13	Check the CSV file corresponding to the test user and chosen language.	The CSV file should reflect the changes to the reviewed words' correct and incorrect counts.	Pass

12. Appendix G – Test Case 6 – Change Settings and View Statistics Test

Test Case ID	TC-6
Test Case Name	Change Settings and View Statistics Test
Brief Description	This test case verifies various changes to the options in the settings menu and that the statistics page is correctly showing the user's statistics.
Priority	Medium
Planned Test Duration	30 minutes
Current Version	2.0
Date of First Version	10/31/2024
Date of Current Version	11/12/2024
Created By	Jacob Poland
Last Update By	Zachary Kao
Approved By	Ryleigh Sibley
The version of Software Under Test	1.0
Test Engineer to Conduct the Test	Garrett Strader
List of Test Support Software	N/A
List of Test Support Hardware	N/A

Test Case 6 Scenario

Test Step Number	Input	Expected	Result (Pass/Fail)
1	Run the application.	The application starts running.	Pass
2	Log in to the application.	The application displays the main menu.	Pass
3	Click on the Settings button.	The application displays the Settings menu as expected.	Pass
4	Click the Apply Changes button.	The application should not respond to the button press. The smrt.log should not contain any logging	Pass

		information about settings being changed.	
5	Adjust the sliders to the following values: KNOWN THRESHOLD to 1. KNOWN DELTA to 0. WALKING WINDOW SIZE to 20. SRS QUEUE SIZE to 10. FOREIGN TO ENGLISH toggle to OFF. LANGUAGE to French.	The sliders should stay bound to the slider and the current value labels should reflect the updated values. The toggle should reflect the updated value. The dropdown box should behave as expected and contain French. The apply button should change to show that it is now interactable.	Pass
6	Click Apply Changes.	The apply changes button should become grayed out again. The application should log the setting changes to smrt.log.	Pass
7	Click the back button.	The application displays the main menu.	Pass
8	Click the Statistics button.	The application should show the Statistics page with the current known words count.	Pass
9	Click the back button.	The application displays the main menu.	Pass
10	Click the Multiple-Choice Flashcard button.	The application displays the multiple-choice flashcard GUI.	Pass
11	Answer a number of questions correctly.	The application displays the correct feedback each time. The application should log that a word has been correctly translated and marked as known and removed from the walking window in smrt.log.	Pass
12	Click the back button.	The application displays the main menu.	Pass
13	Click the Statistics button.	The application displays the Statistics page. The graph and current known words should be updated to reflect the new values.	Pass
14	Exit the application.	The application closes.	Pass
15	Check smrt.log.	The log should contain logging information confirming the previous steps were successful.	Pass